Trust in Inter-Agent Interactions: Definitions, Computational Models, Protocols, and Applications Across Multi-Agent Ecosystems

IEEE Communications Surveys & Tutorials Working Group on Autonomous and Multi-Agent Systems

***Abstract*—The rapid proliferation of autonomous systems—spanning classical symbolic multi-agent systems (MAS), cyber-physical vehicular and unmanned aerial swarms, next-generation 6G edge-native networks, and large language model (LLM)-powered agentic collectives—has elevated inter-agent trust from an auxiliary social heuristic to a foundational engineering requirement. When autonomous agents discover, negotiate, delegate tasks, and execute economic transactions across organizational and security boundaries without continuous human supervision, traditional perimeter-based security and static public-key infrastructures fail to capture runtime behavioral uncertainty, strategic deception, and cognitive fragility. This paper presents a comprehensive, end-to-end survey of trust in inter-agent interactions. First, we trace the conceptual evolution of inter-agent trust across sociological, socio-cognitive, probabilistic, economic, and linguistic-pragmatic paradigms, establishing a formal six-dimensional taxonomy that disambiguates trust from reputation, trustworthiness, dependability, and hard cryptographic security. Second, we formulate and comparatively analyze six core mathematical modeling families: Bayesian sequential inference (including Beta, Dirichlet, TRAVOS, and state-space particle filtering), subjective logic and evidential reasoning, socio-cognitive and multi-source composite architectures (ReGreT and FIRE), spectral and graph-theoretic reputation propagation (EigenTrust, PeerTrust, and PowerTrust), game-theoretic and crypto-economic stake-slashing mechanisms, and machine learning models encompassing multi-agent reinforcement learning (MARL) and internal multi-head attention trust scoring (A-Trust). Third, we examine the paradigm shift introduced by foundation-model and LLM-based multi-agent systems, detailing how prompt infection, sycophancy, hallucination cascades, and sleeper deception invalidate standalone reputation schemes. Fourth, we evaluate emerging inter-agent interoperability and trust protocols—including Model Context Protocol (MCP), Agent-to-Agent (A2A), Agent Network Protocol (ANP), Agent Payments Protocol (AP2), Ethereum ERC-8004, W3C Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs), zero-knowledge machine learning (zkML), and trusted execution environments (TEEs)—and synthesize a risk-calibrated Tiered Hybrid Trust Blueprint (T0–T3). Finally, we catalog classical and agentic attack-defense matrices, review eight representative application domains, and distill critical lessons learned and future research directions toward resilient, verifiable multi-agent ecosystems.**

***Index Terms*—Inter-agent trust, multi-agent systems (MAS), reputation models, Bayesian trust inference, subjective logic, game theory, large language model (LLM) agents, agentic web protocols, decentralized identifiers (DIDs), zero-knowledge proofs.**

Manuscript prepared in *IEEE Communications Surveys & Tutorials* (COMST) format. This comprehensive survey synthesizes theoretical foundations, mathematical models, cryptographic and decentralized identity protocols, foundation-model multi-agent trust architectures, adversarial threat taxonomies, and cross-domain engineering applications of inter-agent trust.

I. Introduction

The architectural evolution of distributed computing and artificial intelligence over the past three decades has progressively shifted decision-making authority from isolated, human-supervised programs to interconnected societies of autonomous agents [1], [2], [23], [90]. In early symbolic Multi-Agent Systems (MAS) of the 1990s and 2000s—governed by Belief-Desire-Intention (BDI) architectures, the Contract Net Protocol (CNP), and Foundation for Intelligent Physical Agents (FIPA) standards—autonomous software entities negotiated task delegation and resource allocation across structured enterprise and electronic marketplace environments [3], [4], [11], [22]. During the 2010s, the proliferation of peer-to-peer (P2P) overlays, Mobile Ad Hoc Networks (MANETs), the Social Internet of Things (SIoT), Vehicle-to-Everything (V2X) networks, and multi-Unmanned Aerial Vehicle (UAV) swarms extended inter-agent interactions into resource-constrained, mobile cyber-physical environments [27], [33], [38], [39]. Most recently, since 2023, the emergence of foundation models and Large Language Models (LLMs) equipped with persistent memory, external tool invocation, and multi-turn planning has catalyzed a transformative third wave: the *Agentic Web* and LLM-based Agentic Multi-Agent Systems (AMAS) [49], [50], [85], [94], [102].

Across all three evolutionary eras, a foundational question governs the viability of decentralized cooperation: *how can an autonomous agent (the trustor) reliably evaluate, verify, and calibrate its reliance on another autonomous agent (the trustee) when operating under incomplete information, strategic self-interest, environmental noise, and adversarial manipulation?* In open multi-agent ecosystems, interacting agents rarely belong to a single administrative domain or share aligned utility functions [23], [26], [43]. Traditional perimeter-based cybersecurity and static cryptographic credentials (such as Domain Name System (DNS) bindings and Transport Layer Security (TLS) certificates) are necessary to authenticate transport endpoints, yet they remain fundamentally insufficient for inter-agent trust [89], [93], [94]. A valid cryptographic signature proves *who* sent a message, but it cannot guarantee that the sending agent possesses the domain competence to execute a delegated workflow, that its internal reasoning is free from hallucination or sycophancy, or that it has not been subverted at runtime via indirect prompt injection or collusive incentives [53], [54], [75], [83].

As illustrated in Fig. 1, the scope of inter-agent trust has

Era I: Symbolic & Cognitive MAS (1994–2007)	Era II: Networked & P2P/IoT MAS (2003–2019)	Era III: Decentralised & DLT/6G MAS (2016–2024)	Era IV: LLM Agentic Web & Hybrid Trust (2023–2026+)
• Formal computational trust (Marsh 1994) • Socio-cognitive BDI delegation (Castelfranchi & Falcone 1998) • Subjective Logic (Jøsang 2001) • ReGreT (2002), BRS (2002), TRAVOS (2006), FIRE (2006)	• Spectral P2P aggregation: EigenTrust (2003), PeerTrust (2004), PowerTrust (2007) • MANET, WSN, SIoT & V2X node/data-centric trust (Cho et al. 2011; Guo et al. 2017) • Sequential Bayesian & Particle Filter SSTM (Liu 2020)	• Smart-contract reputation & consensus (PBFT, PoS, DAG) • Cross-domain wireless & 6G edge/UAV resource trust • Game-theoretic, auction & MARL cooperative trust (COOPER)	• Agent interoperability protocols: MCP, A2A, ANP, AP2, ERC-8004, x402 • W3C DIDs/VCs (BlockA2A, NANDA) & zkML/TEE proofs • LLM-MAS internal attention trust (A-Trust) & GNN topological defense (G-Safeguard) • Tiered Hybrid Trust (T0–T3)

Figure 1. Evolutionary trajectory of inter-agent trust across four paradigms: from symbolic socio-cognitive MAS and P2P/IoT reputation networks to blockchain-anchored 6G systems and the foundation-model Agentic Web.

expanded dramatically alongside the transition from symbolic software agents to billions of autonomous LLM-powered agents transacting on the Agentic Web [88], [90], [91]. On the one hand, classical computational trust and reputation models—such as Marsh's formalism [5], Castelfranchi and Falcone's socio-cognitive delegation theory [11], [12], Jøsang's Beta Reputation System (BRS) and Subjective Logic [16], [17], [19], Sabater and Sierra's ReGreT [20], [21], Teacy et al.'s TRAVOS [25], Huynh et al.'s FIRE [26], and Kamvar et al.'s EigenTrust [27]—established rigorous mathematical machinery for aggregating direct interaction histories and witness referrals. On the other hand, modern multi-agent deployments introduce three disruptive challenges that classical models were never designed to handle: (i) *cognitive and linguistic fragility*, wherein foundation-model agents exhibit sycophancy, nudge-susceptibility, hallucination cascades, and contagious prompt infection across discourse graphs [56], [76], [81], [102]; (ii) *sub-second cross-domain economic autonomy*, requiring machine-speed discovery, verifiable credentials, cryptographic execution proofs, and programmable stake slashing across standards such as Google's Agent-to-Agent (A2A) protocol, Anthropic's Model Context Protocol (MCP), the Agent Payments Protocol (AP2), HTTP x402 micropayments, and Ethereum's ERC-8004 [86]–[88], [91], [94]; and (iii) *physical-digital convergence* in 6G wireless networks, autonomous vehicular platoons, UAV swarms, and embodied robotics, where trust violations translate directly into physical safety hazards or spectrum interference [38], [66], [67].

Furthermore, empirical market and deployment trends underscore why a unified treatment of inter-agent trust is urgently needed. Enterprise adoption of multi-agent orchestration frameworks has accelerated rapidly between 2024 and 2026, with autonomous agents increasingly entrusted to manage cloud infrastructure, execute cross-border supply-chain procurement, coordinate 6G radio access network (RAN) slicing, and synthesize production code [74], [84], [85]. In such open, multi-stakeholder environments, a single unverified trust assumption—such as accepting a peer agent's self-proclaimed capability card without cryptographic attestation or assimilating an unvetted natural-language summary into shared RAG memory—can trigger systemic cascading failures. Motivated by this convergence of multi-agent theory, network security, and foundation-model safety, this survey provides a comprehensive, tutorial-style synthesis of trust in inter-agent interactions. Specifically, reviewing the literature from 1994 through 2026, this paper systematically addresses four fundamental questions: (1) **Definitions & Taxonomy:** How is inter-agent trust formally defined across cognitive, probabilistic, economic, and linguistic dimensions, and how does it differ from reputation, trustworthiness, and cryptographic security? (2) **Mathematical Models:** What mathematical formulations govern trust inference, uncertainty propagation, social aggregation, strategic game equilibria, and neural attention scoring? (3) **Protocols & Architectures:** How do emerging agentic protocols (A2A, MCP, ANP, AP2, ERC-8004), decentralized identifiers (W3C DIDs/VCs), hardware enclaves (TEEs), and zero-knowledge proofs (zkML) combine to form defense-in-depth trust fabrics? (4) **Threats, Applications & Future Roadmap:** How are inter-agent trust mechanisms attacked, defended, and deployed across cyber-physical and digital domains, and what open challenges remain?

A. Related Surveys and Literature Gap

Over the past two decades, a substantial body of survey literature has examined facets of trust, reputation, and multi-agent coordination, as systematically compared in Table I. Foundational MAS surveys by Ramchurn *et al.* [23] and Sabater and Sierra [22] pioneered the classification of individual-level (learning, socio-cognitive, and reputation-based) and system-level (institutional and mechanism-design) trust models for software agents. Jøsang *et al.* [18] broadened the scope to online service provision, contrasting Bayesian, discrete, belief, flow, and fuzzy reputation systems. Subsequent reviews by Pinyol and Sabater-Mir [35], Yu *et al.* [34], Granatyr *et al.* [37], and Braga *et al.* [40] refined socio-cognitive taxonomies and qualitative evaluation dimensions across virtual organizations and electronic marketplaces. In parallel, within the networking and communications community, Cho *et al.* [33] and Guo *et al.* [38] surveyed trust computation models for MANETs and IoT service management, respectively, while Liu [43] unified Beta, Dirichlet, state-space, and subjective logic models under a Generic Bayesian Trust (GBT) perspective.

More recently, the explosion of LLM-powered autonomous agents in 2024–2026 has spurred targeted reviews of agentic security, protocols, and governance. Yu *et al.* [83] proposed the TrustAgent taxonomy, categorizing intrinsic (brain, memory, tool) and extrinsic (agent-to-agent, agent-to-environment, agent-to-user) threats and defenses for LLM agents. Yang *et al.* [84] surveyed AI agent communication protocols—distinguishing context-oriented protocols such as MCP from inter-agent protocols such as A2A, ANP, AITP, and Agora—while Raza *et al.* [85] adapted the AI Trust, Risk, and

Table I
SUMMARY AND COMPARATIVE ANALYSIS OF RELATED SURVEYS ON INTER-AGENT TRUST AND MULTI-AGENT SYSTEMS.

Year		2004	2005	2007	2011	2010	2013	2015	2017	2019	2020	2025	2025	2025	2025	2026	2026
Reference		[23]	[22]	[18]	[33]	[34]	[35]	[37]	[38]	[40]	[43]	[83]	[84]	[85]	[94]	[102]	Our work
Foundations & Definitions	Socio-cognitive & delegation theory	●	●	◐	◐	◐	●	●	○	◐	○	○	○	◐	◐	○	●
	Quantitative & probabilistic axioms	●	◐	●	●	●	○	◐	●	●	●	○	○	◐	◐	◐	●
	Pragmatic & linguistic trust (Gricean)	○	○	○	○	○	○	○	○	○	○	◐	○	◐	◐	●	●
	Multidimensional taxonomy (6D)	◐	●	◐	●	◐	◐	●	●	●	◐	●	◐	●	●	◐	●
Mathematical & Computational Models	Bayesian inference (BDTM/DDTM/GBT)	◐	●	◐	●	○	◐	●	◐	●	○	○	○	○	◐	○	●
	Subjective logic & fuzzy/evidential	●	●	●	●	◐	◐	◐	◐	◐	●	○	○	○	○	○	●
	Socio-cognitive composite (FIRE/ReGreT)	●	◐	○	◐	●	●	○	●	○	○	○	○	◐	○	●	
	Graph/spectral (EigenTrust/PeerTrust)	◐	●	◐	●	○	◐	◐	●	◐	○	○	○	◐	○	●	
	Game theory & crypto-economic stake	○	○	◐	◐	○	○	○	○	○	○	◐	○	●	○	●	
	MARL & attention-based (AI-Trust)	○	○	○	○	○	○	◐	◐	○	◐	○	◐	○	●	●	
Protocols & Verifiable Architectures	Agentic protocols (A2A/MCP/ANP/AP2)	○	○	○	○	○	○	○	○	○	○	●	◐	●	◐	●	
	On-chain registries (ERC-8004/x402)	○	○	○	○	○	○	◐	○	○	○	◐	○	●	○	●	
	Decentralized IDs (W3C DID/VC/NANDA)	○	○	○	○	○	○	○	○	○	○	◐	◐	●	○	●	
	Cryptographic proofs (zkML/TEEs)	○	○	○	○	○	○	○	○	○	○	◐	◐	◐	●	○	●
Threats, Defenses & Applications	Classical Sybil/collusion/whitewashing	◐	●	●	●	○	◐	●	●	◐	○	○	○	●	○	●	
	LLM prompt infection/sycophancy/AiTM	○	○	○	○	○	○	○	○	○	○	●	◐	●	●	●	●
	IoT, V2X, UAV & 6G wireless networks	○	○	●	◐	○	○	●	○	◐	◐	○	○	○	○	●	
	Agentic commerce, code & robotics	●	◐	◐	○	○	○	◐	○	◐	○	●	●	●	●	●	●
	Tiered hybrid blueprint (T0–T3)	○	○	○	○	○	○	○	○	○	○	○	○	◐	●	○	●

● Topics comprehensively discussed. ◐ Topics mentioned but not discussed in detail. ○ Topics not included.

Security Management (TRiSM) framework to LLM-based AMAS, introducing the Component Synergy Score (CSS) and Tool Utilization Efficacy (TUE) metrics. Notably, Hu and Rong [94] conducted a comparative study of six protocol-level trust primitives (Brief, Claim, Proof, Stake, Reputation, and Constraint) across A2A, AP2, and ERC-8004, and He *et al.* [102] formulated a Gricean six-dimensional linguistic trust framework coupled with internal multi-head attention scoring (A-Trust).

However, as revealed in Table I, existing surveys exhibit three collective limitations that motivate our work: (1) **Bifurcation Between Classical Mathematical Theory and Modern Agentic AI:** Classical MAS surveys [18], [22], [23], [37], [43] provide rigorous mathematical formulations of Bayesian, subjective logic, and socio-cognitive models but predate LLM agents, decentralized identifiers (W3C DIDs/VCs), zero-knowledge proofs (zkML), and agentic protocols (A2A/MCP/ERC-8004). Conversely, recent LLM-agent surveys [83]–[85] focus on qualitative prompt security or protocol schemas while omitting the mathematical foundations of sequential trust inference, spectral aggregation, and game-theoretic equilibrium analysis. (2) **Incomplete Coverage of Protocol-Cryptographic-Cognitive Integration:** Prior works rarely analyze how internal neural trust evaluators (such as multi-head attention heads in A-Trust [102] and GNN discourse monitors in G-Safeguard [82]) interact with external cryptographic attestations (DIDs/VCs [93], x402 micropayments [91], and staked validation registries [88]). (3) **Lack of Cross-Domain Synthesis:** Existing reviews isolate either wireless/IoT/vehicular networks [33], [38] or pure software/web agents [84], [94], overlooking the unified design principles required when software agents orchestrate physical 6G edge, UAV, and robotic infrastructure.

B. Strengths and Contributions

Bridging the divide between foundational multi-agent theory, communication network engineering, and modern foundation-model agentic systems, this survey provides a self-contained, mathematically rigorous, and practically actionable reference on inter-agent trust. The principal contributions of this paper are summarized as follows:

- **Unified Conceptual Foundations and Six-Dimensional Taxonomy:** We trace the intellectual lineage of inter-agent trust across sociological, socio-cognitive (BDI delegation), probabilistic, game-theoretic, and pragmatic-linguistic (Gricean) traditions. We formally disambiguate trust from reputation, trustworthiness, reliability, and hard security, establish six formal mathematical axioms of inter-agent trust, and introduce a comprehensive six-dimensional taxonomy.
- **Rigorous Mathematical Formulation of Six Modeling Paradigms:** We explicitly formulate and compare six representative mathematical modeling families for inter-agent trust: (i) Bayesian sequential inference (GBT, BDTM/BRS, DDTM, TRAVOS, and Particle Filter SSTM), (ii) Subjective Logic (SLTM discounting/consensus algebras) and fuzzy/possibilistic model checking, (iii) Socio-cognitive composite models (ReGreT and FIRE), (iv) Graph-theoretic and spectral reputation algorithms (EigenTrust, PeerTrust, PowerTrust), (v) Game-theoretic and crypto-economic stake-slashing optimization, and (vi) Machine learning models including MARL and internal multi-head attention trust scoring (A-Trust).
- **In-Depth Analysis of Foundation-Model and LLM-MAS Trust Dynamics:** We dissect the structural shift from deterministic symbolic agents to stochastic LLM-based Agentic Multi-Agent Systems (AMAS), contrasting intrinsic (brain, memory, tool) and extrinsic (agent-to-agent, environment, user) trustworthiness and analyzing why LLM-specific fragilities—prompt infection, sycophancy, hallucination cascades, sleeper deception, and power-seeking misalignment—render single-layer trust models brittle.
- **Comparative Evaluation of Agentic Web Protocols and Verifiable Architectures:** We systematically evaluate how state-of-the-art protocols—MCP, Google A2A, ANP, AITP, Agora, LMOS, AP2, HTTP x402, Ethereum ERC-8004, W3C DIDs/VCs (BlockA2A, NANDA), zkML, and TEE remote attestations—operationalize the six protocol trust primitives (Brief, Claim, Proof, Stake, Reputation, Constraint), culminating in a risk-calibrated Tiered Hybrid Trust Blueprint (T0–T3).
- **Comprehensive Attack-Defense Matrix and Cross-Domain Applications:** We synthesize both classical reputation attacks (Sybil, collusion, whitewashing, on-off milking) and novel LLM-MAS interaction attacks (cooperative persuasion, Agent-in-the-Middle, CORBA/Prompt Infection contagion) alongside topological, cryptographic, and runtime constraint defenses, and examine eight deployment domains spanning IoT/SIoT, V2X/CAVs, UAV/SAGIN, 6G wireless resource management, agentic commerce, software/scientific workflows, healthcare, and embodied robotics.
- **Critical Lessons Learned and Forward-Looking Research Roadmap:** We identify six fundamental engineering bottlenecks—including the verification-latency paradox, dynamic nudge vulnerability, cross-protocol semantic translation, steganographic multi-agent collusion, privacy-auditability tension, and longitudinal benchmark scarcity—and outline concrete future research directions.

C. Organization and Nomenclature

The organizational structure of this survey is depicted in Fig. 2, and the key acronyms and abbreviations used throughout the manuscript are summarized in Table II. Section II establishes the formal definitions, mathematical axioms, lifecycle architecture, and multidimensional taxonomy of inter-agent trust. Section III presents the mathematical formulations and comparative analysis of the six core computational trust modeling paradigms. Section IV investigates the emerging paradigm of trust in foundation-model and LLM-based multi-agent systems. Section V examines inter-agent communication protocols, decentralized identity frameworks, verifiable execution proofs, and the Tiered Hybrid Trust Blueprint (T0–T3).

Section I	**Introduction:** Historical Evolution (Symbolic MAS → P2P/IoT → DLT/6G → Agentic Web), Related Surveys (Table I), Contributions, and Nomenclature (Table II)
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Section II	**Foundations, Definitions & Multidimensional Taxonomy:** Cross-Disciplinary Definitions, Distinctions (Trust vs. Reputation vs. Security), 6 Axiomatic Properties, Trust Lifecycle (Fig. 3), and 6D Taxonomy (Table III)
⇓	
Section III	**Mathematical & Computational Models:** (A) Bayesian Sequential Inference (GBT, BDTM, DDTM, TRAVOS, SSTM) · (B) Subjective Logic (SLTM) & Fuzzy/Possibilistic · (C) Socio-Cognitive Composite (ReGreT, FIRE) · (D) Graph & Spectral (EigenTrust, PeerTrust) · (E) Game Theory & Crypto-Economic Stake (Table IV) · (F) MARL & Internal Multi-Head Attention (A-Trust) (Table V)
⇓	
Section IV	**Emerging Paradigm—Trust in Foundation-Model & LLM-Based MAS:** Symbolic vs. LLM-MAS Shift · Intrinsic (Brain, Memory, Tool) vs. Extrinsic Trustworthiness (TrustAgent & TRiSM CSS/TUE) · LLM Fragilities (Prompt Infection, Sycophancy, Hallucination, Deception, Misalignment) · Debate, Attention & GNN Topological Defenses
⇓	
Section V	**Inter-Agent Trust Protocols, Decentralized Identity & Verifiable Architectures:** Six Protocol Trust Models (Brief, Claim, Proof, Stake, Reputation, Constraint) · MCP, A2A, ANP, AITP, Agora, LMOS · AP2 Mandates, x402 Micropayments & ERC-8004 · W3C DIDs/VCs (BlockA2A, NANDA) · zkML & TEE Attestations · Tiered Blueprint T0–T3 (Fig. 4, Table VI)
⇓	
Section VI	**Attacks, Vulnerabilities & Defense Mechanisms:** Classical Reputation Attacks (Sybil, Whitewashing, Collusion, On-Off) · Agentic Attacks (Persuasion, AiTM, Prompt Infection, CORBA, Agent Smith) · Defense-in-Depth & Benchmarks (Table VII)
⇓	
Section VII	**Applications & Use Cases:** IoT/SIoT · IoV/V2X/CAVs · Multi-UAV & SAGIN · 6G Wireless Resource Management · Agentic Commerce & DeFi · Software Engineering & Science · Healthcare · Embodied Robotics
⇓	
Section VIII & IX	**Lessons Learned, Open Challenges & Conclusion:** Verification-Latency Paradox, Nudge Fragility, Semantic Interoperability, Steganographic Collusion, Privacy vs. Auditability, Benchmarks (Fig. 5), and Concluding Remarks

Figure 2. Organizational structure and roadmap of this survey.

Section VI details adversarial threat models, contagious interaction attacks, defense-in-depth countermeasures, and evaluation benchmarks. Section VII reviews practical applications across eight cyber-physical, wireless networking, and digital ecosystems. Section VIII synthesizes lessons learned, open challenges, and future research directions, and Section IX concludes the paper.

II. Foundations, Definitions, and Multidimensional Taxonomy of Inter-Agent Trust

Despite its pervasive invocation across computer science, artificial intelligence, and network engineering, the term *trust* is frequently overloaded, conflating psychological beliefs, statistical reliability, social reputation, and cryptographic verification [12], [18], [46], [94]. To establish a rigorous analytical foundation for inter-agent interactions, this section synthesizes formal definitions of trust across five intellectual traditions, delineates the boundaries between trust and adjacent security concepts, formalizes the fundamental mathematical axioms of trust, and presents a unified six-dimensional taxonomy.

A. Conceptual Evolution and Formal Definitions of Inter-Agent Trust

In any inter-agent interaction, a directional relationship is formed between at least two autonomous entities: the **trustor** agent $a_i \in \mathcal{A}$ (the delegating, querying, or relying party) and the **trustee** agent $a_j \in \mathcal{A}$ (the executing, responding, or service-providing party), operating within a specific task, context, or environmental state $c \in \mathcal{C}$ at time step t [5], [23], [43]. Across the literature, formal definitions of inter-agent trust fall into five complementary paradigms:

1) Sociological and Vulnerability-Centric Definitions: Originating in organizational theory and sociology, Gambetta [6] defined trust as the subjective probability with which an agent assesses that another agent or group of agents will perform a particular action, both before the trustor can monitor such action (or independently of its capacity ever to monitor it) and in a context in which it affects the trustor's own action. Mayer, Davis, and Schoorman [7] and Rousseau *et al.* [8] crystallized the vulnerability criterion: trust is a state comprising the intention to accept vulnerability in a situation involving risk, grounded in positive expectations of the competence, benevolence, and integrity of the trustee. Philosophically, O'Neill [9], [10] and Manzini *et al.* [52] emphasize that vulnerability under uncertainty is what separates genuine trust from mere mechanical reliance: one relies on a deterministic calculator, whereas an agent trusts a peer precisely when the peer possesses the autonomy to defect, err, or deceive.

2) Socio-Cognitive and Delegation-Based Definitions: In symbolic and BDI multi-agent systems, Castelfranchi and Falcone [11], [12] formulated the foundational socio-cognitive model of trust as the mental substrate of *delegation*. In their theory, only a cognitive agent endowed with explicit goals and beliefs can trust another agent. Specifically, for trustor a_i to delegate task α to trustee a_j toward achieving goal g_i, the mental attitude of trust $\mathcal{T}^{\text{cog}}(a_i, a_j, \alpha, g_i)$ is constructed from five core belief ingredients: (i) *Competence belief:* a_i believes that a_j possesses the capability, knowledge, and resources required to execute α; (ii) *Disposition/Willingness belief:* a_i believes that a_j is actually willing and committed to performing α as requested; (iii) *Dependence belief:* a_i believes that it either cannot achieve g_i alone or that delegating to a_j yields strictly higher expected utility; (iv) *Fulfillment belief:* a_i expects that a_j will in fact carry out the action appropriately;

Table II
SUMMARY OF KEY ACRONYMS AND ABBREVIATIONS USED THROUGHOUT THIS SURVEY.

Acronym	Definition	Acronym	Definition
A2A	Agent-to-Agent Protocol	LTL	Linear Temporal Logic
AComP / AConP	Agent Communication / Connect Protocol	MANET	Mobile Ad Hoc Network
AiTM	Agent-in-the-Middle Attack	MARL	Multi-Agent Reinforcement Learning
AITP	Agent Interaction & Transaction Protocol	MAS	Multi-Agent System
AMAS	Agentic Multi-Agent System	MCP	Model Context Protocol
ANP	Agent Network Protocol	NANDA	Networked Agents and Decentralized AI
AP2	Agent Payments Protocol	PBFT	Practical Byzantine Fault Tolerance
A-Trust	Attention-Based Trust Score	PF / IPF	Particle Filter / Iterative Particle Filter
ATTR	Advisor-to-Trustor Relevance	PKI	Public Key Infrastructure
BDI	Belief-Desire-Intention Architecture	PoT	Proof-of-Thought Consensus
BDTM	Beta Distribution-Based Trust Model	QoS	Quality of Service
BRS	Beta Reputation System	RAG	Retrieval-Augmented Generation
CAV	Connected and Autonomous Vehicle	SAGIN	Space-Air-Ground Integrated Network
CNP	Contract Net Protocol	SIoT	Social Internet of Things
CORBA	Contagious Recursive Blocking Attack	SLA	Service Level Agreement
CSS	Component Synergy Score	SLTM	Subjective Logic-Based Trust Model
DDTM	Dirichlet Distribution-Based Trust Model	SSTM	State-Space-Based Trust Model
DID	Decentralized Identifier (W3C)	TEE	Trusted Execution Environment
DLT	Distributed Ledger Technology	TMS	Trust Management System
ERC-8004	Ethereum Standard for Trustless Agents	TRAVOS	Trust and Reputation in Virtual Organisations
FIRE	Framework for Integrated Reputation & Trust	TRiSM	Trust, Risk, and Security Management
GBT	Generic Bayesian Trust Framework	TUE	Tool Utilization Efficacy
GNN	Graph Neural Network	UAV	Unmanned Aerial Vehicle
IoV / V2X	Internet of Vehicles / Vehicle-to-Everything	VC / VP	Verifiable Credential / Verifiable Presentation
LMOS	Language Model Operating System	zkML / ZKP	Zero-Knowledge ML / Zero-Knowledge Proof

and (v) *Safety/Benevolence belief:* a_i believes that a_j is non-hostile, permitting a_i to make itself less defended and more vulnerable [12], [35]. Recent work on intelligent AI delegation [99] extends this socio-cognitive view to autonomous AI workflows by coupling capability matching with responsibility assignment, continuous monitoring, and dynamic revocation.

3) Quantitative and Decision-Theoretic Computational Definitions: Marsh [5] introduced the first formal mathematical representation of computational trust in MAS, modeling trust as a continuous scalar variable over the half-open interval $[-1, +1)$, where -1 denotes complete distrust, 0 represents complete uncertainty or neutrality, and values approaching $+1$ signify maximal trust. Marsh decomposed trust into three levels: *basic trust* T_i (the trustor's general predisposition to trust derived from all past experiences), *general trust* $T_i(j)$ (the trustor's overall evaluation of trustee a_j independent of a specific situation), and *situational trust* $T_i(j, \alpha)$, defined as:

$$T_i(j, \alpha) = U_i(\alpha) \cdot I_i(\alpha) \cdot \widehat{T_i(j)}, \tag{1}$$

where $U_i(\alpha) \in [-1, +1]$ denotes the utility gained by a_i from situation α, $I_i(\alpha) \in [0, 1]$ represents the subjective importance of situation α, and $\widehat{T_i(j)}$ is an estimate of general trust tempered by optimism or pessimism. Cooperation occurs if and only if situational trust exceeds a risk-dependent threshold:

$$T_i(j, \alpha) > \Theta_i(j, \alpha) = \frac{\text{Perceived_Risk}_i(j, \alpha)}{\text{Perceived_Competence}_i(j, \alpha) + \widehat{T_i(j)}} \cdot I_i(\alpha). \tag{2}$$

This decision-theoretic formulation directly links trust assessment to action selection under risk.

4) Probabilistic and Epistemic Uncertainty Definitions: Because point-estimate trust scores conflate neutral evidence with complete ignorance, Jøsang [16], [18] and Liu [43] define inter-agent trust as a second-order probability distribution or epistemic belief state over a latent parameter $\Theta_{i,j}^{(c)} \in [0, 1]$ representing the probability that trustee a_j will fulfill its commitment in context c. Under this paradigm, trust is not merely a scalar expectation $\mathbb{E}[\Theta_{i,j}^{(c)}]$ but the full posterior density $p(\theta_{i,j,k}^{(c)} \mid y_{1:k})$ conditioned on direct interaction outcomes and third-party testimonies $y_{1:k}$, or equivalently an opinion tuple $\omega_j^i = (b, d, u, a)$ that explicitly quantifies belief b, disbelief d, epistemic uncertainty u, and base rate a [19].

5) Linguistic, Pragmatic, and Protocol-Centric Definitions in Agentic AI: In LLM-based Multi-Agent Systems (LLM-MAS), agents interact primarily through natural language dialogues, structured JSON schemas (e.g., AgentCards), and tool invocations [86], [94], [102]. Drawing upon Grice's cooperative principle in human pragmatics [45], He *et al.* [102] define inter-agent message trustworthiness across six orthogonal linguistic dimensions: (1) *Factual accuracy* (verifiable truth supported by evidence), (2) *Logical consistency* (internal coherence free from contradictions or fallacies), (3) *Contextual relevance* (direct alignment with the task prompt and discourse state), (4) *Objectivity/Bias-freedom* (absence of ideological, emotional, or manipulative skew), (5) *Clarity* (unambiguous, concise formulation), and (6) *Language quality* (syntactic and stylistic integrity). Complementarily, at the protocol infrastructure layer, Hu and Rong [94] define inter-agent trust as a composite protocol guarantee engineered through six composable primitives: *Brief* (third-party verifiable credentials), *Claim* (self-asserted capabilities), *Proof* (cryptographic or hardware attestations), *Stake* (slashable economic collateral), *Reputation* (distributed social feedback), and *Constraint*

(sandboxed capability bounding).

6) Disambiguating Trust, Trustworthiness, Reputation, Dependability, and Security: To maintain analytical precision throughout this survey, we formally distinguish five closely related yet distinct concepts in multi-agent systems [10], [18], [22], [85], [94]:

Trust (Subjective Directional Attitude): The subjective, context-specific belief, expectation, or epistemic distribution held by trustor a_i regarding the future behavior of trustee a_j for task c, accompanied by a willingness to accept vulnerability under risk.

Trustworthiness (Objective Intrinsic Property): An objective property of the trustee a_j reflecting its actual competence, alignment, safety, and integrity in domain c [10], [85]. The central objective of any trust management system is *trust calibration*—aligning subjective trust $T_{i\to j}^{(c)}$ with objective trustworthiness $\Theta_j^{*(c)}$ so as to minimize both *over-trust* (delegating to an untrustworthy or compromised agent) and *under-trust* (rejecting a capable, honest collaborator).

Reputation (Collective Social Perception): A public or community-aggregated perception of trustee a_j synthesized from historical referrals, witness testimonies, and interaction ratings across a network of agents [13], [21], [27]. Reputation is one input signal to trust ($T_{i\to j} = f(\text{Direct}_{i,j}, \text{Rep}_j, \text{Proof}_j, \dots)$), not synonymous with trust itself: an agent may distrust a globally reputable peer due to private negative experience or runtime anomaly detection.

Dependability and Reliability (Non-Adversarial Fault Tolerance): Classical systems attributes measuring mean time between failures, availability, and deterministic correctness against random hardware or channel faults. Unlike reliability, trust explicitly accounts for strategic self-interest, goal misalignment, cognitive manipulation, and intentional deception.

Hard Security and Access Control (Cryptographic Enforcement): Mechanisms such as encryption, digital signatures, and role-based access control (RBAC) that prevent unauthorized external access [33], [38]. While hard security protects the communication pipe and verifies identity, soft/computational trust evaluates whether an authenticated insider or peer will act beneficially once granted access.

B. Fundamental Mathematical Properties and Axioms of Inter-Agent Trust

Any mathematically sound computational trust model for inter-agent interactions must account for six fundamental structural properties observed across empirical and theoretical studies [5], [18], [33], [43], [94]:

Axiom 1 — Directional Asymmetry: Trust is inherently non-symmetric. For any two interacting agents $a_i, a_j \in \mathcal{A}$, $T(a_i \to a_j, c, t) \neq T(a_j \to a_i, c, t)$ in general [43]. Even when observing the exact same interaction outcome y_k, a_i and a_j possess distinct prior beliefs, risk tolerances, and subjective likelihood functions $p_{i\to j}(y_k \mid \theta_j) \neq p_{j\to i}(y_k \mid \theta_i)$.

Axiom 2 — Context and Task Multi-Facetedness: Trust is scoped to specific capability domains. High trust in agent a_j for natural-language summarization (c_1) does not imply trust in a_j for executing privileged financial transactions (c_2) or cooperative spectrum sensing (c_3). Cross-context transfer requires explicit semantic similarity weighting $\text{Sim}(c_1, c_2) \in [0, 1]$ [22], [23].

Axiom 3 — Attenuated and Conditional Transitivity: Trust is not strictly transitive in the algebraic sense, yet it supports *discounted referral propagation* along social paths [19], [36]. If a_i trusts advisor a_k regarding its referral honesty ($T^{\text{rec}}(a_i \to a_k)$) and a_k trusts functional provider a_j ($T^{\text{func}}(a_k \to a_j)$), the derived indirect trust satisfies the attenuation inequality $T^{\text{ind}}(a_i \to a_k \to a_j) \leq \min\{T^{\text{rec}}(a_i \to a_k), T^{\text{func}}(a_k \to a_j)\}$, with epistemic uncertainty monotonically increasing with hop count.

Axiom 4 — Dynamic Non-Monotonicity and Asymmetric Temporal Decay: Trust evolves continuously over time and exhibits *hysteresis* (the slow-to-build, fast-to-destroy principle) [32], [33]. Specifically, the marginal trust gain from a cooperative outcome is strictly smaller than the marginal trust penalty inflicted by a defection or safety violation: $|\Delta T^{+}(y_k = 1)| \ll |\Delta T^{-}(y_k = 0)|$. Furthermore, in the absence of interactions, trust decays exponentially toward an epistemic prior or ignorance state at rate $\lambda \in (0, 1)$.

Axiom 5 — Epistemic Uncertainty vs. Distrust Separation: Lack of evidence ($u = 1$) must be mathematically distinguished from balanced conflicting evidence ($b = d = 0.5, u = 0$) [16], [25]. A newly onboarded agent (*cold-start state*) carries high epistemic uncertainty, requiring exploratory probing or third-party verifiable credentials rather than punitive blacklisting.

Axiom 6 — Runtime Non-Accumulation under Cognitive/Policy Drift: In foundation-model and LLM-based multi-agent systems, historical reputation does not guarantee runtime safety if the agent's prompt context is poisoned, its underlying model weights are updated, or the transaction stakes exceed its accumulated reputation capital [94], [102]. High-impact actions must therefore enforce *zero-trust verification* independent of historical standing.

C. Six-Dimensional Taxonomy of Inter-Agent Trust

As illustrated in the closed-loop lifecycle of Fig. 3, inter-agent trust mechanisms operate across multiple phases and layers of the system stack. To systematically organize the vast literature, we classify inter-agent trust frameworks along six orthogonal dimensions, summarized in Table III:

Dimension 1 — Origin of Trust Evidence: Categorizes where trust signals originate: (a) *Direct / Interaction Trust* derived from first-hand bilateral outcomes [5], [17]; (b) *Indirect / Witness Reputation* gathered from peer referrals and transitive social graphs [20], [25], [27]; (c) *Socio-Structural / Role & Stereotype Trust* inferred from institutional norms, organizational roles, or feature-based stereotypes prior to direct interaction [26], [31]; (d) *Certified / Credential Briefs* issued by trusted third-party authorities or decentralized ledgers [89], [93]; and (e) *Intrinsic Neural / Discourse Signals* extracted directly from an LLM agent's internal hidden states or multi-head attention activations upon receiving a message [102].

Dimension 2 — Protocol Assurance Mechanism: Following the protocol design framework of Hu and Rong [94], encompasses *Claim* (unverified self-descriptions), *Brief* (verifiable endorsements/VCs), *Reputation* (community feedback

Phase 1: Bootstrapping & Discovery		**Phase 2: Multi-Source Evidence Collection**		**Phase 3: Trust Computation & Risk Calibration**		**Phase 4: Execution, Audit & Feedback Update**
• Self-Assertions / Claims (A2A AgentCards, MCP schemas) • Identity & Briefs (W3C DIDs, Verifiable Credentials, NANDA) • On-Chain Discovery (ERC-8004 Identity Registry)	$\xrightarrow{\text{Filter}}$	• Direct Experience History $\mathcal{H}_{i,j}$ • Witness Referrals & Social Graph (T_{WR}, EigenTrust) • Internal LLM Signals (Multi-Head Attention $\text{Attn}^{l,h}$) • Cryptographic Proofs (TEE Attestation, zkML, Mandates)	$\xrightarrow{\text{Fuse}}$	• Bayesian / Subjective Logic / FIRE / Spectral Aggregation • Attention Trust Score (A-Trust) & GNN Discourse Check • Tier Selection (T0–T3) & Stake-to-Risk Calibration	$\xrightarrow{\text{Gate}}$	• Action Policy: *Accept*, *Question/Verify*, or *Reject* • Sandboxed Execution (Constraints, LTL Pruning) • Asymmetric Decay, Reputation Update & Stake Slashing

Figure 3. End-to-end closed-loop lifecycle architecture of inter-agent trust management across bootstrapping, evidence collection, trust computation, and policy enforcement.

scores), *Stake* (bonded economic collateral with slashing), *Proof* (cryptographic signatures, ZKPs, TEE remote attestations), and *Constraint* (execution sandboxes, least-privilege capabilities, and formal temporal logic guardrails).

Dimension 3 — Mathematical Modeling Paradigm: Spans (a) *Probabilistic & Bayesian Inference* (GBT, BDTM, DDTM, SSTM), (b) *Belief, Evidential & Fuzzy Logics* (Subjective Logic, Dempster-Shafer, Fuzzy/Possibilistic CTL), (c) *Socio-Cognitive & Heuristic Composite* (Castelfranchi-Falcone BDI, ReGreT, FIRE), (d) *Graph & Spectral Flow* (EigenTrust, PeerTrust, PowerTrust, GNNs), (e) *Game-Theoretic & Mechanism Design* (repeated games, signaling games, VCG auctions, staking), and (f) *Machine Learning & Foundation-Model Evaluators* (MARL, A-Trust, LLM-as-a-Judge debate).

Dimension 4 — Assessment Granularity and Target: Distinguishes *Message/Data-Centric Trust* (evaluating whether a specific incoming statement or sensor packet is truthful and consistent [38], [102]), *Action/Tool-Centric Trust* (authorizing a specific API call, payment mandate, or physical actuation [67], [87]), *Entity/Node-Centric Trust* (maintaining a persistent longitudinal profile of an agent's reliability), and *Coalition/System-Level Trust* (evaluating emergent safety and synergy across an entire multi-agent topology [82], [85]).

Dimension 5 — Architectural Topology: Covers *Centralized / Hub-and-Spoke* registries (platform-managed directories and orchestrators), *Decentralized P2P / Distributed Hash Table (DHT)* overlays (Chord/CAN-based local witness storage [27], [28]), *Hierarchical / Federated Edge-Cloud* architectures (cloudlet/RSU/UAV cluster heads [39]), and *Ledger-Anchored / On-Chain Registry* architectures (smart contracts, DIDs, ERC-8004 [88], [91]).

Dimension 6 — Temporal Phase in the Interaction Lifecycle: Classifies mechanisms into *Ex-Ante (Pre-Interaction)* gating (discovery, credential verification, stake bonding), *In-Situ (Runtime)* monitoring (attention anomaly detection, sandboxed execution, multi-agent debate), and *Ex-Post (Post-Interaction)* settlement (Bayesian posterior updating, reputation propagation, cryptographic audit verification, and collateral slashing).

III. Mathematical and Computational Models of Inter-Agent Trust

To translate the conceptual definitions and axioms of Section II into executable algorithms for autonomous agents, researchers have developed a rich hierarchy of mathematical trust and reputation models. Mirroring the analytical structure of classical and contemporary multi-agent systems, we categorize computational trust models into six mathematical families: (1) Bayesian and sequential probabilistic inference models, (2) Subjective logic and fuzzy/evidential reasoning models, (3) Socio-cognitive and multi-source integrated frameworks, (4) Graph-theoretic and spectral reputation propagation algorithms, (5) Game-theoretic, auction, and crypto-economic stake-slashing models, and (6) Machine learning, Multi-Agent Reinforcement Learning (MARL), and internal multi-head attention models. Table V provides a comprehensive synthesis of these six modeling families.

A. Bayesian and Sequential Probabilistic Trust Models

Because inter-agent trust is inherently uncertain and subjective, Bayesian probability theory provides a natural mathematical apparatus for updating a trustor's belief distribution as interaction outcomes and third-party testimonies arrive sequentially [17], [25], [43]. As formalized in Liu's Generic Bayesian Trust (GBT) framework [43], let the latent random variable $\Theta_k \in [0,1]$ (with realization θ_k) denote the probabilistic trustworthiness of trustee a_j from the perspective of trustor a_i at discrete time step k, and let $y_{1:k} \triangleq \{y_1, y_2, \ldots, y_k\}$ denote the sequence of observed interaction outcomes and witness reports up to step k. Sequential Bayesian trust inference alternates between a *prediction stage* (governed by the Chapman-Kolmogorov equation) and an *update stage* (governed by Bayes' rule):

$$\text{Transition Law:} \quad \theta_k = f_k\left(\theta_{k-1}, v_{k-1}\right), \tag{3a}$$

$$\text{Prediction Stage:} \quad p\left(\theta_k \mid y_{1:k-1}\right) = \int p\left(\theta_k \mid \theta_{k-1}\right) p\left(\theta_{k-1} \mid y_{1:k-1}\right) d \tag{3b}$$

$$\text{Posterior Update:} \quad p\left(\theta_k \mid y_{1:k}\right) = \frac{p\left(y_k \mid \theta_k\right) p\left(\theta_k \mid y_{1:k-1}\right)}{\int p\left(y_k \mid \tilde{\theta}_k\right) p\left(\tilde{\theta}_k \mid y_{1:k-1}\right) d\tilde{\theta}_k}, \tag{3c}$$

$$\text{Particle Weighting:} \quad \omega_k^{(s)} \propto \omega_{k-1}^{(s)} \frac{p\left(\theta_k^{(s)} \mid \theta_{k-1}^{(s)}\right) p\left(y_k \mid \theta_k^{(s)}\right)}{q\left(\theta_k^{(s)} \mid \theta_{k-1}^{(s)}, y_k\right)}, \quad \sum_{s=1}^{N_s} \omega_k^{(s)} \tag{3d}$$

In (3a), $f_k(\cdot)$ characterizes the temporal evolution of the trustee's behavior and v_{k-1} is an independent and identically distributed (i.i.d.) process noise sequence modeling behavioral drift. When non-conjugate likelihoods $p(y_k \mid \theta_k)$ or nonlinear state transitions preclude closed-form integration in (3c), a Particle Filter (PF) approximates the posterior density via weighted trajectories $\{\theta_{0:k}^{(s)}, \omega_k^{(s)}\}_{s=1}^{N_s}$ drawn from proposal distribution $q(\cdot)$ as shown in (3d) [43]. Major classical Bayesian

Table III
MULTIDIMENSIONAL TAXONOMY OF TRUST IN INTER-AGENT INTERACTIONS.

Dimension	Category / Sub-Type	Core Mechanism & Analytical Scope	Representative Systems	Refs.
1. Evidence Origin	Direct / Interaction (T_{IT})	First-hand bilateral task outcomes, SLA compliance, latency, and quality ratings	Marsh, BRS, ReGreT, FIRE	[5], [17], [21], [26]
	Indirect / Witness (T_{WR})	Third-party referrals, social network recommendations, and transitive chains	TRAVOS, EigenTrust, PeerTrust	[25], [27], [28]
	Role & Stereotype (T_{RT})	Prior expectations from organizational roles, institutional norms, or agent profiles	FIRE, Stereotype-Trust, BDI	[12], [26], [31]
	Certified Briefs (T_{CR})	Tamper-proof cryptographic attestations, W3C DIDs/VCs, and third-party certificates	FIRE T_{CR}, DID/VC, NANDA	[26], [89], [93]
	Intrinsic Neural Signals	Internal multi-head attention activations and discourse consistency in LLM backbones	A-Trust, Attention Tracker	[65], [102]
2. Protocol Assurance	Claim & Brief	Self-declared AgentCards/schemas vs. issuer-signed verifiable credentials	A2A, MCP, ANP, BlockA2A	[84], [86], [94]
	Proof & Constraint	Zero-knowledge proofs (zkML), TEE enclaves, signed Mandates, and LTL sandboxes	AP2, ERC-8004, SELP	[67], [87], [88]
	Stake & Reputation	Slashable crypto-economic collateral bonds and distributed feedback registries	ERC-8004, x402, EigenTrust	[88], [91], [94]
3. Granularity	Message & Data Level	Truthfulness, logical consistency, relevance, clarity, and sensor packet validity	A-Trust, Data-Centric IoV	[38], [102]
	Action & Tool Level	Per-transaction authorization, API capability bounding, and payment mandates	AP2 Mandates, ToolGuard	[83], [87]
	Agent & Coalition Level	Longitudinal agent reliability and system-wide Component Synergy Score (CSS)	FIRE, TRiSM, G-Safeguard	[26], [82], [85]
4. Architecture	Centralized / Orchestrator	Single directory, cloud fusion center, or enterprise API gateway	Enterprise A2A, BRS	[17], [86]
	Decentralized P2P / Edge	DHT overlays, fog/cloudlet hierarchies, and local witness graphs	PeerTrust, SecuredTrust, BC-Trust	[28], [32], [39]
	Ledger-Anchored / DLT	On-chain identity/reputation/validation registries and DID resolvers	ERC-8004, Hyperledger Indy, x402	[88], [91], [93]

trust models emerge as closed-form or state-space specializations of (3a)–(3d):

1) Beta Distribution-Based Trust Model (BDTM) and Beta Reputation System (BRS): When interaction outcomes are binary ($y \in \{\text{cooperative/success}, \text{defection/failure}\}$) and the state transition is static ($\theta_k = \theta_{k-1}$), the binomial likelihood is conjugate to the Beta prior $\text{Beta}(\alpha, \beta)$, yielding Jøsang and Ismail's Beta Reputation System (BRS) [17]. After observing $r_{i,j}$ positive outcomes and $s_{i,j}$ negative outcomes between trustor a_i and trustee a_j (with uniform prior $\alpha_0 = \beta_0 = 1$), the posterior density and point-estimate trust score with exponential forgetting factor $\lambda \in (0, 1]$ over K observation epochs are given by:

$$p\left(\theta \mid \alpha_{i,j}, \beta_{i,j}\right) = \frac{\Gamma\left(\alpha_{i,j} + \beta_{i,j}\right)}{\Gamma\left(\alpha_{i,j}\right)\Gamma\left(\beta_{i,j}\right)} \theta^{\alpha_{i,j}-1}\left(1-\theta\right)^{\beta_{i,j}-1}, \tag{4a}$$

$$\alpha_{i,j}^{(K)} = 1 + \sum_{k=1}^{K} \lambda^{K-k} r_{i,j}^{(k)}, \qquad \beta_{i,j}^{(K)} = 1 + \sum_{k=1}^{K} \lambda^{K-k} s_{i,j}^{(k)}, \tag{4b}$$

$$T_{i,j}^{\text{BDTM}} = \mathbb{E}\left[\Theta \mid \alpha_{i,j}^{(K)}, \beta_{i,j}^{(K)}\right] = \frac{\alpha_{i,j}^{(K)}}{\alpha_{i,j}^{(K)} + \beta_{i,j}^{(K)}}, \qquad \text{Var}[\Theta] = \frac{\alpha_{i,j}^{(K)}\beta_{i,j}^{(K)}}{\left(\alpha_{i,j}^{(K)} + \beta_{i,j}^{(K)}\right)^2\left(\alpha_{i,j}^{(K)} + \beta_{i,j}^{(K)} + 1\right)}. \tag{4c}$$

To incorporate witness reports from a third-party advisor a_w that reports evidence $(r_{w,j}, s_{w,j})$, Liu and Yang [43] weight the advisor's counts by the Advisor-to-Trustor Relevance (ATTR) or advisor trust score $\theta_{i,w} \in [0, 1]$, updating the posterior parameters via $\alpha_{i,j} \leftarrow \alpha_{i,j} + \theta_{i,w} r_{w,j}$ and $\beta_{i,j} \leftarrow \beta_{i,j} + \theta_{i,w} s_{w,j}$.

2) TRAVOS (Confidence-Interval Witness Filtering): Rather than assuming homogeneous witness honesty as in basic BRS, Teacy *et al.*'s TRAVOS (Trust and Reputation model for Agent-based Virtual OrganisationS) [25] computes an explicit epistemic confidence metric $\gamma_{i,j} \in [0, 1]$ representing the probability that the actual parameter θ lies within an acceptable error tolerance ϵ around the point estimate $T_{i,j}^{\text{BDTM}}$:

$$\gamma_{i,j} = \frac{\int_{T_{i,j}^{\text{BDTM}}-\epsilon}^{T_{i,j}^{\text{BDTM}}+\epsilon} \theta^{\alpha_{i,j}-1}\left(1-\theta\right)^{\beta_{i,j}-1} d\theta}{\int_0^1 \theta^{\alpha_{i,j}-1}\left(1-\theta\right)^{\beta_{i,j}-1} d\theta}. \tag{5}$$

Only when direct confidence $\gamma_{i,j}$ falls below a predetermined threshold θ^{γ} does trustor a_i solicit witness opinions. Furthermore, TRAVOS discounts unfair or collusive witnesses by binning historical advisor reports $(\hat{\alpha}_{w,j}, \hat{\beta}_{w,j})$ and evaluating the empirical accuracy of advisor a_w against a_i's subsequent first-hand observations.

3) Dirichlet Distribution (DDTM) and State-Space Trust Models (SSTM): To capture multi-graded interaction quality (e.g., b discrete service levels from catastrophic failure to exemplary fulfillment), Dirichlet Distribution-Based Trust Models (DDTM) [18], [43] represent trust as a vector $\boldsymbol{\Theta} = [\Theta_1, \ldots, \Theta_b]$ on the $(b-1)$-simplex governed by conjugate Dirichlet parameters $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_b)$. After observing outcome count vector $\boldsymbol{n} = (n_1, \ldots, n_b)$, the posterior is $\text{Dir}(\alpha_1 + n_1, \ldots, \alpha_b + n_b)$ with expected class probabilities $\mathbb{E}[\Theta_m] = (\alpha_m + n_m)/\sum_{\ell=1}^{b}(\alpha_\ell + n_\ell)$. Conversely, for continuous-valued observations in dynamic wireless sensor

and IoT environments, the State-Space Trust Model (SSTM) [43] models temporal drift explicitly via a truncated Gaussian transition density and neighbor-voting likelihood:

$$\theta_k \sim \mathcal{T}_{\mathcal{N}}\left(\alpha_{\text{decay}}\theta_{k-1}, Q; [0,1]\right), \tag{6a}$$

$$p\left(y_k \mid \theta_k\right) = \exp\left(-\frac{|\theta_k - V_k|}{\beta_{\text{sens}}}\right), \qquad V_k = \frac{\sum_{\ell=1}^{n_k} \theta_{k,\ell}\, \mathbb{I}\left(|y_{k,\ell} - \theta_{k,0}| \leq r_{\text{tol}}\right)}{\sum_{\ell=1}^{n_k} \theta_{k,\ell}} \tag{6b}$$

where $\alpha_{\text{decay}} \in [0,1]$ (typically 0.85) is the state persistence factor, Q is the process variance, V_k is the trust-weighted spatial agreement vote among n_k neighboring peers within tolerance r_{tol}, and Iterative Particle Filtering (IPF) jointly estimates $\boldsymbol{\theta}_k = [\theta_{k,0}, \theta_{k,1}, \ldots, \theta_{k,n_k}]$.

B. Subjective Logic and Fuzzy/Evidential Reasoning Models

Jøsang's Subjective Logic Trust Model (SLTM) [16], [19] operates directly on second-order epistemic opinions to preserve explicit ignorance during transitive trust propagation and multi-source fusion. A binomial subjective opinion held by agent a_i regarding proposition or trustee a_j is a quadruple $\omega_j^i = (b_j^i, d_j^i, u_j^i, a_j^i)$, where $b_j^i, d_j^i, u_j^i \in [0,1]$ denote belief, disbelief, and epistemic uncertainty (ignorance), respectively, satisfying the simplex additivity constraint, and $a_j^i \in [0,1]$ denotes the prior base rate:

$$b_j^i + d_j^i + u_j^i = 1, \qquad P\left(\omega_j^i\right) = b_j^i + a_j^i u_j^i, \tag{7a}$$

$$\left(b_j^i, d_j^i, u_j^i\right) = \left(\frac{r_{i,j}}{r_{i,j} + s_{i,j} + W}, \frac{s_{i,j}}{r_{i,j} + s_{i,j} + W}, \frac{W}{r_{i,j} + s_{i,j} + W}\right), \tag{7b}$$

where $P(\omega_j^i)$ in (7a) is the projected probability expectation and (7b) defines the bijective mapping between Beta/Dirichlet evidence counts $(r_{i,j}, s_{i,j})$ and subjective opinions with non-informative prior weight $W = 2$ [19], [43]. Subjective logic equips agents with two fundamental algebraic operators for trust network analysis: (i) the *transitive discounting operator* ($\otimes$), where a_i discounts advisor a_k's functional opinion ω_j^k by its referral opinion ω_k^i; and (ii) the *cumulative consensus fusion operator* ($\oplus$), which combines independent opinions ω_j^i and ω_j^k (for $u_j^i + u_j^k - u_j^i u_j^k \neq 0$):

$$\omega_j^{i:k} = \omega_k^i \otimes \omega_j^k = \left(b_k^i b_j^k,\ b_k^i d_j^k,\ d_k^i + u_k^i + b_k^i u_j^k,\ a_j^k\right), \tag{8a}$$

$$\omega_j^{i \diamond k} = \omega_j^i \oplus \omega_j^k = \left(\frac{b_j^i u_j^k + b_j^k u_j^i}{\kappa},\ \frac{d_j^i u_j^k + d_j^k u_j^i}{\kappa},\ \frac{u_j^i u_j^k}{\kappa},\ \frac{a_j^i u_j^k + a_j^k u_j^i - \left(a_j^i + a_j^k\right) u_j^i u_j^k}{\kappa - u_j^i u_j^k}\right) \tag{8b}$$

where $\kappa = u_j^i + u_j^k - u_j^i u_j^k$. Note how (8a) mathematically enforces Axiom 3 (attenuated transitivity): any disbelief d_k^i or uncertainty u_k^i toward the advisor is mapped directly into epistemic uncertainty $u_j^{i:k}$ regarding the target trustee, rather than falsely inflating disbelief $d_j^{i:k}$.

Complementing subjective logic, fuzzy set theory and possibilistic model checking capture linguistic vagueness in contract enactment and behavioral verification [23], [29], [100]. Ramchurn *et al.* [23] model contract issues (e.g., price, delivery latency, quality) using fuzzy membership functions $\mu_L(x) \in [0,1]$ over linguistic labels $L \in \mathcal{L}$, deriving expected utility loss intervals via the intersection of personal confidence and social reputation ranges. More recently, formal verification of multi-agent trust under fuzzy uncertainty has been formulated via Generalized Possibilistic Trust Interpreted Systems (GPTIS) and Generalized Possibilistic Computation Tree Logic (GPTCTL) [100], enabling automated model checking of quantitative possibility bounds $\|\Phi\|(s) \in [0,1]$ for temporal trust reachability formulas across state transitions.

C. Socio-Cognitive and Multi-Source Integrated Models (ReGreT and FIRE)

In open multi-agent environments where direct interaction history may be sparse or non-existent, composite architectures fuse heterogeneous evidence streams. Sabater and Sierra's ReGreT system [20], [21] integrates three dimensions: (i) *Direct Trust*, computed as a recency-weighted mean of local outcome impressions $r_k \in [-1,+1]$ accompanied by a rating deviation reliability metric; (ii) *Witness Reputation*, which uses fuzzy rules over sociograms (competition, cooperation, and trade social networks) to estimate witness mendacity before aggregating referrals; and (iii) *Neighborhood and System Reputation*, which assign default expectations based on an agent's local social clique and institutional role.

Extending ReGreT, Huynh, Jennings, and Shadbolt proposed FIRE (an integrated trust and reputation model for open multi-agent systems) [26], which synthesizes four distinct trust sources for trustor a_i evaluating trustee a_j with respect to contract term c: (1) **Interaction Trust** (T_{IT}) from direct experience; (2) **Role-Based Trust** (T_{RT}) from domain-specific rule bases encoding institutional norms; (3) **Witness Reputation** (T_{WR}) from referral chains within a branching factor and hop limit; and (4) **Certified Reputation** (T_{CR}), wherein trustee a_j proactively presents cryptographically signed rating certificates from past partners during initial contact, solving the cold-start problem without multi-hop network flooding. For each component $K \in \{\text{IT}, \text{RT}, \text{WR}, \text{CR}\}$ backed by rating set $\mathcal{R}_K(a_i, a_j, c)$, FIRE computes the component score T_K, sample-size reliability ρ_{NK}, deviation reliability ρ_{DK}, and composite trust T_{FIRE} as:

$$T_K\left(a_i, a_j, c\right) = \frac{\sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right) \cdot v\left(r_\ell\right)}{\sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right)}, \qquad \omega_{\text{IT}}\left(r_\ell\right) = e^{-\frac{\Delta t(r_\ell)}{\tau_{\text{rec}}}}, \tag{9a}$$

$$\rho_{NK}\left(a_i, a_j, c\right) = \begin{cases} \sin\left(\frac{\pi}{2\gamma_K} \sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right)\right), & \text{if } \sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right) \leq \\ 1, & \text{otherwise,} \end{cases} \tag{9b}$$

$$\rho_{DK}\left(a_i, a_j, c\right) = 1 - \frac{1}{2}\sqrt{\frac{\sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right)\left(v\left(r_\ell\right) - T_K\left(a_i, a_j, c\right)\right)^2}{\sum_{r_\ell \in \mathcal{R}_K} \omega_K\left(r_\ell\right)}}, \tag{9c}$$

$$T_{\text{FIRE}}\left(a_i, a_j, c\right) = \frac{\sum_{K \in \{\text{IT,RT,WR,CR}\}} W_K \cdot \rho_K \cdot T_K\left(a_i, a_j, c\right)}{\sum_{K \in \{\text{IT,RT,WR,CR}\}} W_K \cdot \rho_K}, \qquad \rho_K = \tag{9d}$$

where $v(r_\ell) \in [-1,+1]$ is the rating value, γ_K is the intimacy threshold of rating cardinality, and W_K are user-

configurable or adaptive structural weights. Notably, FIRE's Certified Reputation (T_{CR}) directly anticipates modern W3C Verifiable Credentials and NANDA AgentFacts [89], [93], [94] by shifting testimony storage to the trustee while relying on public-key signatures to prevent fabrication (though still requiring cherry-picking defenses).

D. Graph-Theoretic and Spectral Reputation Propagation Models

When millions of agents interact over decentralized P2P overlays or service meshes, local bilateral evidence must be aggregated into a global or personalized reputation vector over a directed interaction graph $\mathcal{G} = (\mathcal{A}, \mathcal{E})$. Kamvar, Schlosser, and Garcia-Molina's seminal **EigenTrust** algorithm [27] defines the raw local trust $s_{i,j} = \text{sat}(i,j) - \text{unsat}(i,j)$ from binary transaction ratings $\{-1, +1\}$, truncates negative values to zero, and row-normalizes them into a stochastic transition matrix $\mathbf{C} = [c_{i,j}]$. To break malicious collusive cliques and Sybil rings, EigenTrust anchors the random-surfer walk in a prior distribution $\mathbf{p}$ over a set of *pre-trusted peers* $\mathcal{P}$, computing the global reputation vector $\mathbf{t}$ via distributed power iteration:

$$c_{i,j} = \frac{\max(s_{i,j}, 0)}{\sum_{m \in \mathcal{A}} \max(s_{i,m}, 0)}, \qquad \mathbf{t}^{(k+1)} = (1 - a_{\text{pre}})\, \mathbf{C}^T \mathbf{t}^{(k)} + a_{\text{pre}} \mathbf{p}, \tag{10a}$$

where $a_{\text{pre}} \in (0,1)$ is the reset probability toward pre-trusted anchors. Because static pre-trusted peers in EigenTrust can penalize honest newcomers or become central bottlenecks [47], Zhou and Hwang's **PowerTrust** [30] dynamically elects the highest-reputation power nodes over a power-law overlay using a lookahead random walk.

To defend against collusive ballot-stuffing, slandering, and oscillating behavior without relying solely on pre-trusted anchors, Xiong and Liu's **PeerTrust** [28], [41] incorporates five factors into the trust evaluation of agent u over its $I(u)$ transactions:

$$T_{\text{PeerTrust}}(u) = \alpha_{\text{pt}} \sum_{i=1}^{I(u)} S(u,i) \cdot \text{Cr}(p(u,i)) \cdot \text{TF}(u,i) + \beta_{\text{pt}}\, \text{CF}(u), \tag{10b}$$

where $S(u,i) \in [0,1]$ is the normalized satisfaction rating for transaction i, $p(u,i)$ is the reporting peer, $\text{Cr}(p(u,i))$ is the feedback credibility weight of the rater (computed either via Trust-Value Mode PeerTrust$_{\text{TVM}}$ or Personalized Similarity Mode PeerTrust$_{\text{PSM}}$ using root-mean-square rating similarity over co-rated peers), $\text{TF}(u,i)$ is the transaction context factor (e.g., monetary value or task criticality), and $\text{CF}(u)$ is the community context factor incentivizing feedback submission [28], [41]. Furthermore, Das and Islam's **SecuredTrust** [32] extends PeerTrust to dynamic multi-agent service balancing by combining recent trust, historical trust, expected trust deviation against strategic on-off oscillations, non-linear decay, and heuristic load-balancing across reputable providers.

E. Game-Theoretic, Auction, and Crypto-Economic Stake Models

While statistical reputation models treat partner behavior as a stochastic process to be estimated, game-theoretic and mechanism-design models treat autonomous agents as strategic utility maximizers whose honesty must be incentivized as a Nash, Bayesian-Nash, Stackelberg, or evolutionary equilibrium [23], [24], [94]. Table IV summarizes the primary game-theoretic and economic paradigms applied to inter-agent trust.

In open agentic economies—such as Ethereum ERC-8004 [88] and ledger-anchored A2A/x402 markets [91], [94]—where pseudonymous identities are cheap and agents may interact only once, repeated-game reputation alone cannot deter *last-mile betrayal*. The **Stake and Slashing** model resolves this by requiring trustee a_j to lock economic collateral $S_j \geq 0$ in a smart contract or escrow prior to task execution. Let R_j denote the service fee paid upon valid completion, C_{exec} the honest computational cost, G_{defect} the illicit gain obtainable by cheating or returning a fabricated output, $p_{\text{detect}}(\phi) \in [0,1]$ the probability that verification mechanism ϕ (e.g., staked re-execution, TEE attestation, or zk-proof verification) detects misbehavior, and ΔV_{rep} the discounted future value of lost reputation. The trustee's expected utilities under honest execution ($a_j = \text{H}$) and strategic defection ($a_j = \text{D}$), and the resulting incentive-compatibility constraint for trustless cooperation, are formulated as:

$$U_j(\text{H}) = R_j - C_{\text{exec}}, \tag{11a}$$

$$U_j(\text{D}) = (1 - p_{\text{detect}}(\phi))(R_j + G_{\text{defect}}) - p_{\text{detect}}(\phi)(S \tag{11b}$$

$$\text{Incentive Compatibility:} \quad S_j \geq \frac{(1 - p_{\text{detect}}(\phi))\, G_{\text{defect}} + C_{\text{exec}} - p_{\text{detect}}(\phi} {p_{\text{detect}}(\phi)} \tag{11c}$$

Equation (11c) reveals a fundamental design law of inter-agent protocols: *Proof, Stake, and Reputation are economic substitutes* [94]. When cryptographic verification is deterministic ($p_{\text{detect}}(\phi) \to 1$ via ZKP or TEE), minimal stake is required; when verification relies on probabilistic spot-auditing ($p_{\text{detect}}(\phi) \ll 1$) and the agent lacks established reputation ($\Delta V_{\text{rep}} \approx 0$), the required bond S_j scales inversely with $p_{\text{detect}}(\phi)$.

F. Machine Learning, MARL, and Internal Multi-Head Attention Models

In high-dimensional, non-stationary multi-agent environments, hand-crafted heuristic weights often fail to capture complex nonlinear correlations among behavioral features. Machine learning approaches address this across two paradigms: (i) **Multi-Agent Reinforcement Learning (MARL) with Emergent Reputation** [96], [97], where decentralized agents jointly learn an action policy $\pi_i(a_i \mid o_i, \hat{\mathbf{r}}_{-i})$ and a reputation assignment function $g_i(\hat{r}_{i \to j} \mid \tau_j)$ over local observation trajectories (such as the decentralized COOPER algorithm [96] and Multi-Agent Trust Region Policy Optimization MA-TRPO [97]), stabilizing cooperation in social dilemmas without a central coordinator; and (ii) **Internal Multi-Head Attention Trust Scoring (A-Trust)** for LLM-based multi-agent systems, recently introduced by He *et al.* [102].

Specifically, in LLM-MAS, prompting an external LLM verifier to judge every inter-agent message incurs prohibitive

Table IV
SUMMARY OF GAME-THEORETIC, AUCTION, AND CRYPTO-ECONOMIC MODELS IN INTER-AGENT TRUST MANAGEMENT.

Game / Mechanism Type	Player 1 / Leader	Player 2 / Follower	Strategic Structure & Incentive Condition	Solution Concept	Refs.
Repeated Trust & Delegation Game	Trustor a_i (Delegator)	Trustee a_j (Executor)	Honesty sustained via folk-theorem Grim-Trigger or Tit-for-Tat when discount factor $\delta \geq \frac{G_{\text{cheat}}}{G_{\text{cheat}}+V_{\text{future}}}$	Subgame Perfect Nash Equilibrium (SPNE)	[6], [23]
Stackelberg Incentive & Slashing Game	Protocol / Verifier (Leader)	Service Agent a_j (Follower)	Leader sets collateral bond S_j, audit rate p_{audit}, and reward R_j; Follower chooses effort/honesty $e_j \in [0,1]$	Stackelberg Equilibrium (Backward Induction)	[88], [94]
Bayesian Signaling & Screening Game	Sender Agent (Private Type θ_j)	Receiver Agent (Verifier)	High-competence type θ_H posts costly credential/stake signal m_j that low type θ_L cannot profitably mimic	Perfect Bayesian Equilibrium (Separating)	[24], [93]
Evolutionary Coalition Trust Game	Bounded-Rational Agent Population	Peer Sub-Populations	Replicator dynamics over cooperative, defective, and trust-selective strategies in dependence networks	Evolutionarily Stable Strategy (ESS)	[44], [101]
Trust-Weighted VCG / Double Auction	Task Auctioneer / Orchestrator	Bidding Worker Agents	Social welfare maximization weighting bid valuations b_j by verified trust score $T_{i,j}$ with truthful payment rule	Dominant-Strategy Incentive Compatible (DSIC)	[24], [91]
Adversarial Attacker-Defender Game	Defender MAS (LLAMOS / Filter)	Compromised / Colluding Agents	Minimax zero-sum or general-sum game over prompt filtering, topology pruning, and attack injection budgets	Mixed-Strategy Nash / Stackelberg Equilibrium	[82], [83]

token latency and suffers from the verifier's own hallucinations. Instead, He *et al.* [102] discovered that when an LLM backbone F with L layers and H attention heads per layer processes a concatenated input prompt $P = C \parallel M$—where C denotes the local agent's system profile, task instructions, and dialogue history, and M denotes an incoming message of token length $|M|$ from a peer agent—the internal self-attention weights assigned from the final token of P to the tokens of M exhibit specialized activation signatures whenever M violates one of the six Gricean trust dimensions $\mathcal{D} = \{\text{fact}, \text{logic}, \text{rel}, \text{bias}, \text{clar}, \text{qual}\}$. Letting $\alpha_t^{l,h} \in (0,1)$ denote the attention weight from the last token to the t-th token of message M in layer $l \in \{1,\dots,L\}$ and head $h \in \{1,\dots,H\}$, A-Trust computes the geometric-mean token attention score, averages across layers to obtain an H-dimensional head feature vector $\mathbf{Attn}(M)$, and evaluates dimension-specific logistic classifiers $f_r(\cdot)$ as follows:

$$\text{Attn}^{l,h}(M) = \prod_{t=1}^{|M|} \left(\alpha_t^{l,h}\right)^{\frac{1}{|M|}} = \exp\left(\frac{1}{|M|}\sum_{t=1}^{|M|} \ln \alpha_t^{l,h}\right), \tag{12a}$$

$$\text{Attn}^{h}(M) = \frac{1}{L}\sum_{l=1}^{L} \text{Attn}^{l,h}(M), \qquad \mathbf{Attn}(M) = \left[\text{Attn}^{1}(M), \dots, \text{Attn}^{H}(M)\right]^T \in \mathbb{R}^H, \tag{12b}$$

$$s_r(M) = f_r(\mathbf{Attn}(M)) = \sigma\left(\mathbf{w}_r^T \mathbf{Attn}(M) + b_r\right), \quad \forall r \in \mathcal{D}, \tag{12c}$$

$$S_{\text{A-Trust}}(M) = \sum_{r\in\mathcal{D}} \eta_r s_r(M), \qquad R_{i,j}^{(k)} = \gamma_{\text{mem}} R_{i,j}^{(k-1)} + (1-\gamma_{\text{mem}}) S_{\text{A-Trust}}\left(M_{j\to i}^{(k)}\right), \tag{12d}$$

where $\sigma(\cdot)$ is the logistic sigmoid, η_r are task-dependent dimension weights, $S_{\text{A-Trust}}(M)$ is the message-level trustworthiness score, and $R_{i,j}^{(k)}$ in (12d) is the dynamic agent-level trust record maintained by the Trust Management System (TMS) with memory factor $\gamma_{\text{mem}} \in [0,1]$ [102]. Because $\alpha_t^{l,h}$ is already computed during the standard forward pass of the receiving LLM, A-Trust achieves near-zero latency overhead while outperforming external LLM judges against subtle communication attacks.

IV. EMERGING PARADIGM: TRUST IN FOUNDATION-MODEL AND LLM-BASED MULTI-AGENT SYSTEMS

The integration of Large Language Models (LLMs) and multimodal foundation models as the cognitive core of autonomous agents has fundamentally altered the landscape of multi-agent systems [49], [50], [83], [85], [94]. Whereas classical MAS relied on deterministic rule engines, structured FIPA-ACL performatives, and pre-defined ontologies, LLM-based Multi-Agent Systems (LLM-MAS)—such as AutoGen, MetaGPT, CAMEL, CrewAI, LangGraph, and GPTSwarm—communicate in open-ended natural language, synthesize executable code on the fly, query external retrieval-augmented generation (RAG) memories, and invoke privileged APIs across organizational boundaries. This section examines why foundation-model agents invalidate core assumptions of classical trust models and details the emerging architectural frameworks designed to restore trustworthiness.

A. Structural Shift from Symbolic MAS to LLM-Powered Agentic Collectives

In classical BDI architectures (e.g., Jason, JADE), an agent's internal beliefs, desires, and intentions are discrete symbolic predicates, and its action space is bounded by a finite library of pre-verified plans [2], [4]. Under that regime, if an agent was non-malicious and its software was free of bugs, its behavior was stationary and predictable, allowing reputation systems like BRS, TRAVOS, and FIRE to converge reliably to the agent's true competence [17], [25], [26]. By contrast, LLM-based Agentic Multi-Agent Systems (AMAS) exhibit four structural departures:

Stochastic and Non-Stationary Reasoning: Autoregressive token sampling introduces inherent output variance across

Table V
COMPREHENSIVE COMPARISON OF REPRESENTATIVE MATHEMATICAL AND COMPUTATIONAL TRUST MODELS IN INTER-AGENT SYSTEMS.

Model	Mathematical Family	Input / Space	Core Formulation	Complexity	Primary Limitation	Refs.
BDTM / BRS	Bayesian Conjugate	Binary $\{0,1\}$	Beta(α,β); $E[\Theta]=\frac{\alpha}{\alpha+\beta}$	$\mathcal{O}(1)$ update; analytic conjugate	Vulnerable to cold-start & whitewashing without witnesses	[17], [43]
TRAVOS	Bayesian + Confidence	Binary $\{0,1\}$	Integral confidence $\gamma_{i,j}$ + advisor accuracy binning	$\mathcal{O}(N_{\text{adv}})$ when $\gamma_{i,j}<\theta^{\gamma}$	Requires extensive repeated history with each advisor	[25]
DDTM	Multinomial Bayesian	Discrete b-ary	Dir$(\alpha_1,\ldots,\alpha_b)$ simplex posterior	$\mathcal{O}(b)$ analytic update	Static state assumption unless coupled with forgetting factor	[18], [43]
SSTM / GBT-PF	State-Space / Particle Filter	Continuous $\mathbb{R}^n$	$\theta_k \sim \mathcal{T}_{\mathcal{N}}(\alpha\theta_{k-1}, Q)$ + IPF voting	$\mathcal{O}(N_s \cdot n_k)$ per particle step	Higher computational cost for large particle sets N_s	[43]
Subjective Logic (SLTM)	Second-Order Belief Algebra	Triplet (b,d,u,a)	Discounting $\otimes$ and Consensus $\oplus$ operators	$\mathcal{O}(\lvert E\rvert)$ along series-parallel paths	Correlated witness paths require graph canonicalization	[16], [19]
ReGreT & FIRE	Socio-Cognitive Composite	Continuous $[-1,+1]$	Weighted fusion of $T_{\text{IT}}, T_{\text{RT}}, T_{\text{WR}}, T_{\text{CR}}$ via $\rho_{NK}\rho_{DK}$	$\mathcal{O}(d^{\text{hop}})$ witness query or $\mathcal{O}(1)$ via T_{CR}	Rule-base tuning for T_{RT}; certificate cherry-picking in T_{CR}	[21], [26]
EigenTrust & PowerTrust	Spectral Graph / Markov Walk	Normalized Matrix $\mathbf{C}$	$\mathbf{t}^{(k+1)}=(1-a)\mathbf{C}^T\mathbf{t}^{(k)}+a\mathbf{p}$	$\mathcal{O}(I_{\text{iter}}\cdot\lvert E\rvert)$ distributed DHT	Relies on pre-trusted peers $\mathcal{P}$ or power-node election	[27], [30]
PeerTrust & SecuredTrust	Credibility-Weighted P2P	Multiscale $[0,1]$	5-factor $S(u,i)\text{Cr}(p)\text{TF}(u,i)+\text{CF}(u)$ + decay	$\mathcal{O}(I(u)\cdot\lvert\text{CoRated}\rvert)$	Communication overhead for personalized similarity PSM	[28], [32]
Crypto-Economic Stake	Game Theory / Mechanism	Verifiable Fault ϕ	Collateral S_j slashing when $p_{\text{detect}}S_j > G_{\text{defect}}$	$\mathcal{O}(1)$ escrow + verifier cost	Ex-post penalty; capital barrier for resource-poor agents	[88], [94]
A-Trust (LLM-MAS)	Internal Attention Classifier	Token Stream $P=C\parallel M$	Geometric token attention $\text{Attn}^{l,h}(M)$ + 6D logistic	$\mathcal{O}(L\cdot H\cdot\lvert M\rvert)$ (free in forward pass)	Requires white-box access to receiving LLM attention weights	[102]

identical prompts, complicating deterministic auditability and reproducibility [85].

Unstructured Natural-Language Control Plane: Because data and instructions share the same token stream (the context window), every incoming inter-agent message acts simultaneously as passive information and as an executable control prompt capable of altering the receiver's policy [53], [102].

Implicit Over-Trust Bias: Instruction-tuned and RLHF-aligned LLMs are trained to be helpful and cooperative toward their prompt context; consequently, vanilla LLM agents treat all incoming peer messages as authoritative instructions without epistemic skepticism [76], [102].

Open-Ended Tool and Environment Actuation: Modern agents dynamically compose shell commands, database queries, web browser actions, and financial transactions, greatly amplifying the blast radius of a single compromised inter-agent interaction [83], [87].

B. Intrinsic vs. Extrinsic Trustworthiness and Quantitative AMAS Metrics

To systematically deconstruct trustworthiness in LLM-MAS, Yu *et al.*'s **TrustAgent** framework [83] partitions agentic trustworthiness into two coupled spheres: (1) **Intrinsic Trustworthiness**, encompassing the internal reliability of the *Brain* (the foundation model's alignment, truthfulness, and resistance to jailbreaks), *Memory* (the integrity and privacy of short-term context and long-term vector RAG stores against poisoning and membership inference), and *Tools* (safe parameter validation and sandboxed invocation of external APIs); and (2) **Extrinsic Trustworthiness**, governing *Agent-to-Agent Interaction* (cooperation, competition, and multi-round debate), *Agent-to-Environment Interaction* (physical robotics/AVs and digital web/finance/healthcare environments), and *Agent-to-User Interaction* (human-in-the-loop trust calibration and explainability).

Complementing TrustAgent, Raza *et al.* [85] adapted the AI Trust, Risk, and Security Management (**TRiSM**) lifecycle to AMAS across four pillars—Explainability, ModelOps, Application Security/Privacy, and Governance—and introduced two system-level quantitative metrics to evaluate inter-agent operational trustworthiness across an N-agent workflow: the **Component Synergy Score (CSS)** and **Tool Utilization Efficacy (TUE)**:

$$\text{CSS}=\frac{1}{|\mathcal{E}_{\text{collab}}|}\sum_{(a_i,a_j)\in\mathcal{E}_{\text{collab}}}\left(\omega_{\text{succ}}\cdot\text{SR}_{i\to j}+\omega_{\text{coh}}\cdot\text{SemCoh}\left(M_{i\to j}\right)-\omega_{\text{co}}\right. \tag{13a}$$

$$\text{TUE}=\frac{\sum_{k=1}^{K_{\text{tool}}}\mathbb{I}\left(\text{ValidCall}_k\wedge\text{PolicyCompliant}_k\wedge\text{TaskGain}_k>0\right)}{K_{\text{tool}}+\lambda_{\text{pen}}\sum_{k=1}^{K_{\text{tool}}}\mathbb{I}\left(\text{UnauthorizedOrRedundant}_k\right)}, \tag{13b}$$

where $\text{SR}_{i\to j}$ denotes the handoff task success rate between

collaborating agents $(a_i, a_j) \in \mathcal{E}_{\text{collab}}$, $\text{SemCoh}(M_{i\to j})$ measures semantic coherence and protocol compliance of inter-agent messages, $\text{CR}_{i,j}$ quantifies inter-agent state conflict or loop oscillation rates, and $\text{TUE} \in [0,1]$ penalizes hallucinated, redundant, or policy-violating tool invocations across K_{tool} calls [85].

C. Five LLM-Specific Fragilities That Undermine Classical Trust Models

A critical finding of recent comparative studies [83], [94], [102] is that five cognitive fragilities unique to foundation models invalidate the core assumptions of standalone Claim or Reputation systems:

Indirect Prompt Injection and Contagious Prompt Infection: Because LLMs cannot inherently separate untrusted peer data from system instructions, an adversarial message from a single compromised agent or external retrieval source can hijack a benign agent's control flow [53], [56]. Consequently, an agent with a pristine historical reputation ($T_{i,j}^{\text{BDTM}} \approx 1$) can be subverted instantaneously at runtime into acting as a confused deputy or self-propagating attack node.

Hypersensitivity to Nudging and Sycophancy: RLHF-trained LLMs systematically exhibit sycophantic behavior—adapting their factual assertions and reasoning to agree with confident or authoritative interlocutors [75]. Cherep *et al.* [76] demonstrated that LLM agents are hypersensitive to subtle conversational nudges, allowing a persuasive adversarial agent in a multi-agent debate to gradually shift a benign agent's beliefs and objectives without triggering explicit jailbreak filters [59].

Confident Hallucinations and Information Cascades: LLMs generate fluent, syntactically polished outputs that may be factually fabricated [77], [78]. In multi-agent workflows, a hallucinated intermediate claim produced by an upstream planner or researcher agent is often accepted uncritically by downstream verifier and executor agents, inducing a *hallucination cascade* across the MAS topology [61].

Strategic Deception and Sleeper Agents: Advanced foundation models can exhibit strategic deception—behaving cooperatively during evaluation or low-stakes monitoring phases to accumulate high reputation scores, and defecting only when a specific trigger condition or high-value transaction arises [54], [95]. Hubinger *et al.* [54] showed that such backdoor deceptive policies can persist through standard safety fine-tuning.

Emergent Power-Seeking and Insider Misalignment: When endowed with long-horizon planning and autonomous tool access, misaligned agents may pursue instrumental subgoals—such as acquiring unauthorized API credentials, resisting shutdown, or monopolizing compute resources—turning trusted internal collaborators into insider threats [55], [79].

D. Collaborative, Attention-Based, and Topological Trust Defenses for LLM-MAS

To counteract these foundation-model fragilities, three classes of LLM-native trust mechanisms have emerged:

Multi-Agent Debate, Proof-of-Thought, and Cross-Auditing: Rather than trusting a single agent's output, collaborative defense frameworks employ structured multi-agent verification. *BlockAgents* [62] integrates multi-round LLM debate voting with a Byzantine-resilient Proof-of-Thought (PoT) consensus mechanism to identify and down-weight Byzantine peers. *Audit-LLM* [63] and *AutoDefense* [64] deploy specialized auditor and moderator agents that decompose peer responses into verifiable atomic claims and filter adversarial prompts before they reach the executor agent.

Attention-Based Trust Management Systems (A-Trust TMS): Building on the A-Trust score $S_{\text{A-Trust}}(M)$ and agent trust record $R_{i,j}^{(k)}$ defined in (12a)–(12d), He *et al.* [102] implement a three-action policy controller at each receiving agent: (i) *Accept* if both message score $S_{\text{A-Trust}}(M) \geq \tau_{\text{high}}$ and sender record $R_{i,j}^{(k)} \geq \tau_{\text{agent}}$; (ii) *Question / Challenge* if $\tau_{\text{low}} \leq S_{\text{A-Trust}}(M) < \tau_{\text{high}}$, prompting the sender to supply step-by-step evidence on the specific violated Gricean dimension (e.g., factual accuracy or logical consistency); and (iii) *Reject & Quarantine* if $S_{\text{A-Trust}}(M) < \tau_{\text{low}}$, pruning the message from the context window and penalizing $R_{i,j}^{(k)}$.

Topological Optimization and GNN Discourse Anomaly Detection: Modeling the multi-agent system as a directed temporal discourse graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{X}_t)$ enables structural containment of contagious attacks. *GPTSwarm* [51] optimizes inter-agent communication edges via reinforcement learning to prune unreliable links, while *G-Safeguard* [82] applies Graph Neural Networks (GNNs) over multi-turn agent dialogue embeddings to detect anomalous persuasion and infection subgraphs across chain, tree, star, and fully connected MAS topologies, isolating compromised nodes before contagion reaches consensus.

V. Inter-Agent Trust Protocols, Decentralized Identity, and Verifiable Architectures

As autonomous agents transition from closed laboratory sandboxes to an open *Agentic Web* [84], [90], [94]—discovering peers, negotiating contracts, and settling payments across organizational boundaries—trust shifts from ad hoc application logic to standardized network protocols. This section analyzes the six fundamental trust primitives of inter-agent protocol design, evaluates how state-of-the-art protocols (MCP, A2A, ANP, AP2, HTTP x402, and Ethereum ERC-8004) implement these primitives, examines decentralized identity (W3C DIDs/VCs) and cryptographic verification (zkML and TEEs), and synthesizes a risk-calibrated Tiered Hybrid Trust Blueprint (T0–T3).

A. The Six Protocol-Level Trust Models: Brief, Claim, Proof, Stake, Reputation, and Constraint

Synthesizing historical security primitives with contemporary agentic web specifications, Hu and Rong [94] established that all inter-agent protocol trust mechanisms decompose into six foundational models:

Claim (Self-Asserted Identity and Capability Profiles): The agent publishes a machine-readable manifest—such as an *AgentCard* in Google's A2A protocol [86] or an *agents.json* / MCP tool descriptor [84]—declaring its capabilities, supported schemas, endpoint URLs, and operating policies. Claim is lightweight and indispensable for dynamic discovery, but

offers zero standalone assurance against mendacity, prompt injection, or hallucinated competence.

Brief (Verifiable Third-Party Endorsements and Credentials): Trust is bootstrapped via cryptographically signed attestations issued by recognized authorities, auditors, or decentralized registries—exemplified by W3C Verifiable Credentials (VCs) [93], X.509/mTLS certificates, and NANDA Verified AgentFacts [89]. Briefs prevent impersonation and unverified capability inflation, though they remain static between renewal epochs and depend on issuer integrity and timely revocation.

Proof (Cryptographic and Hardware-Attested Evidence): Promises are replaced by mathematically verifiable execution evidence, including digital signatures over tamper-evident action logs, Zero-Knowledge Proofs of correct inference (zkML), and remote attestations from Trusted Execution Environments (TEEs) [88], [94]. Proofs provide trust-minimized verification of integrity and policy compliance even in hostile environments, at the cost of proof-generation latency and circuit/hardware complexity.

Stake (Bonded Collateral, Slashing, and Insurance): Agents post financial or token collateral subject to programmatic slashing if on-chain verifiers, challenge-response protocols, or arbitration quorums detect contract violations [88], [94]. Stake aligns economic incentives and deters Sybil and deceptive behaviors, provided fault conditions are objectively verifiable.

Reputation (Distributed Feedback and Social Graph Signals): Interaction outcomes, SLA adherence, and peer reviews are aggregated into queryable longitudinal scores or spectral rankings [27], [28], [88]. Reputation captures nuanced, subjective quality and long-term drift, but acts as a lagging indicator vulnerable to reputation milking, collusion, and runtime prompt hijacking.

Constraint (Sandboxing, Least Privilege, and Formal Capability Bounding): Rather than trusting the agent's internal disposition, the runtime environment restricts the agent's action space via OS/container sandboxes, scoped ephemeral OAuth/API tokens, rate limiters, payment caps, and Linear Temporal Logic (LTL) action pruning [67], [87], [94]. Constraint bounds the maximum blast radius of any single failure or betrayal.

B. Standardized Agent Interoperability, Payment, and Trustless Registry Protocols

Between 2024 and 2026, industry consortia, open-source foundations, and blockchain standards bodies introduced a suite of protocols to standardize inter-agent interaction and trust, as compared in Table VI:

1) Model Context Protocol (MCP) and Context-Oriented Interfaces: Introduced by Anthropic [84], MCP standardizes the client-server JSON-RPC interface between an LLM host/agent and external context sources, databases, and executable tools. In trust terms, MCP relies primarily on *Claim* (tool and resource descriptors) and host-enforced *Constraint* (local user approval prompts and process isolation). However, native MCP lacks decentralized agent identity, cryptographic proof of tool execution integrity, or cross-domain reputation, leaving agents vulnerable to tool poisoning and malicious MCP servers unless augmented by verifiable layers [93].

2) Google's Agent-to-Agent (A2A) Protocol and Agent Network Protocol (ANP): Google's A2A specification [86], [94] enables multi-turn task collaboration across heterogeneous agent frameworks. Each agent exposes a JSON *AgentCard* at a well-known HTTPS endpoint advertising its identity, version, skills, input/output modalities, and authentication schemes (e.g., OAuth 2.0, OpenID Connect, mTLS). A2A natively combines *Claim* (AgentCards), transport-level *Brief* (TLS/OAuth tokens), and enterprise *Constraint* (API gateways and allowlists), excelling in intra-enterprise and federated B2B orchestration. However, because standard A2A AgentCards are self-asserted and lack protocol-native global reputation, staking, or execution proofs, researchers have proposed **Agent Network Protocol (ANP)** [84] and **BlockA2A** [92], which anchor AgentCards to W3C Decentralized Identifiers (did:wba or ledger DIDs) and append Ed25519 DID signatures (SG-A2A / VG-A2A) to every inter-agent message header.

3) Agent Payments Protocol (AP2) and HTTP x402 Micropayments: When autonomous agents execute commercial transactions on behalf of human or institutional principals, financial accountability becomes paramount. The **Agent Payments Protocol (AP2)** [87], [94] introduces the cryptographic *Mandate*—a signed verifiable credential capturing explicit user authorization, spending caps, itemized cart hashes, and whether a human was present in the loop (human-present vs. human-not-present flows). AP2 operationalizes *Brief* + *Proof* + *Constraint*: agents never handle raw credit card credentials (relying on scoped payment tokenization), and every payment carries a non-repudiable cryptographic audit trail for dispute resolution. Complementarily, for machine-speed, low-value inter-agent API compensation, Vaziry *et al.* [91] integrated the **HTTP 402 (x402 / L402)** micropayment standard with A2A and smart-contract AgentCard registries, allowing a service agent to return an HTTP 402 Payment Required challenge with pricing metadata and settle stablecoin or Lightning micropayments atomically per request.

4) Ethereum ERC-8004 (Trustless Agents) and NANDA Index: To enable open, permissionless agent economies without pre-existing bilateral trust, Ethereum's **ERC-8004 (Trustless Agents)** standard [88], [94] defines three composable on-chain smart contract registries: (1) an *Identity Registry* minting portable ERC-721 non-fungible token (NFT) handles that resolve to off-chain AgentCard URIs; (2) a *Reputation Registry* recording structured, queryable client feedback hashes and scores; and (3) a *Validation Registry* coordinating third-party verification via three pluggable trust modalities—crypto-economic staked re-execution (*Stake*), TEE remote attestation (*Proof*), and zero-knowledge machine learning proofs (*Proof*). In parallel, Raskar *et al.*'s **NANDA** architecture [89] moves beyond static DNS by introducing a decentralized index of *Verified AgentFacts*—cryptographically attested metadata bundles certifying an agent's provenance, safety benchmarks, and capability compliance.

C. Decentralized Identifiers (W3C DIDs), Verifiable Credentials, and Cryptographic Proofs

A critical architectural insight demonstrated by Rodriguez Garzon *et al.* [93] is that equipping AI agents with self-

Tier	Risk Profile & Example Tasks	Mandatory Protocol Trust Primitives	Verification & Identity Requirements	Containment & Audit Posture
T3	**Life / Safety-Critical** Autonomous surgery, grid/nuclear control, CAV platooning, high-value treasury	**Brief + Proof + Stake + Constraint + Human-in-the-Loop** (Multi-layer defense-in-depth)	Regulatory W3C VCs bound to legal entities; multi-agent Byzantine quorum + TEE/ZKP proof; institutional insurance	Non-overridable LTL hard limits; physical fail-safes; real-time immutable audit logs
T2	**High Stakes** Large B2B payments, production code deployment, sensitive medical/legal RAG	**Brief + Proof + Stake + Constraint** (*"Verify relentlessly"*; Reputation ranks candidates only)	Persistent DID + domain VCs; per-action TEE attestation, zkML proof, or ERC-8004 staked validator re-execution	Deny-by-default time-boxed sandbox; slashable bond $S_j \geq$ MaxLoss; signed receipts
T1	**Moderate Stakes** Small reversible payments (AP2/x402), calendar/email writes, standard API calls	**Brief + Claim + Reputation + Soft Stake + Constraint**	Authenticated DID/OAuth + signed AP2 Mandate; minimum reputation threshold $R_{i,j} \geq \tau_{\min}$; micro-bond	Scoped ephemeral tokens; A-Trust message screening; durable append-only receipts
T0	**Low Stakes** Public discovery, read-only queries, brainstorming, non-critical summarization	**Claim + Optional Brief + Soft Constraint** (Frictionless interoperability)	Self-described A2A AgentCard / MCP schema; pseudonym or basic TLS certificate tolerated	Read-only sandbox; rate limits & circuit breakers; best-effort logging

Figure 4. The Tiered Hybrid Trust Blueprint (T0–T3) for inter-agent protocol design, calibrating cryptographic proofs, economic stake, credentials, reputation, and sandbox constraints to task criticality.

sovereign W3C Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) enables spontaneous cross-domain trust bootstrapping at the onset of a dialogue, *provided cryptographic operations are strictly decoupled from the LLM's stochastic token generation*. In their architecture, each agent holds a ledger-anchored DID (resolved via a Decentralized Identity Foundation (DIF) Universal Resolver over Hyperledger Indy / BCovrin) along with base Verifiable Credentials (bVCs) and relationship Verifiable Credentials (rVCs) formatted in JSON-LD, normalized via URDNA2015, and signed with Ed25519 JSON Web Signatures (JWS). Because an LLM cannot reliably compute or verify cryptographic signatures in-context without hallucinating, VC/VP creation and verification are exposed as external deterministic tools via MCP or function-calling adapters [93].

For high-assurance runtime verification (*Proof*), two cryptographic technologies complement DIDs/VCs: (1) **Trusted Execution Environments (TEEs)** (e.g., Intel SGX/TDX, AMD SEV-SNP, and NVIDIA H100 Confidential Computing), which isolate agent code and model weights inside hardware-encrypted memory enclaves and generate cryptographic remote attestation quotes proving that a specific uncorrupted model binary processed a specific input; and (2) **Zero-Knowledge Proofs (ZKPs) and zkML** (e.g., EZKL, Halo2, Groth16), which allow an agent to generate a succinct non-interactive argument of knowledge (zk-SNARK) proving that an output $y = F_{\mathbf{W}}(x)$ or policy check was computed faithfully from committed weights $\mathbf{W}$ and private inputs x without disclosing proprietary prompts or weights [88], [94].

D. The Tiered Hybrid Trust Blueprint (T0–T3)

Because no single trust mechanism suffices across all operational contexts—pure reputation is vulnerable to sycophancy and Sybil attacks, while universal ZKP/TEE verification imposes prohibitive latency on routine queries—modern agentic architectures adopt a **Tiered Hybrid Trust Blueprint (T0–T3)** [94], illustrated in Fig. 4. Under this architecture, two invariants hold across all tiers: (i) *least privilege at the capability boundary* (agents receive only narrow, time-boxed permissions), and (ii) *evidence-first accountability at the audit boundary* (all state-mutating actions produce signed receipts). As task blast radius escalates from read-only discovery (T0) and reversible micro-transactions (T1) to high-stakes financial/infrastructure execution (T2) and safety-critical physical control (T3), the protocol gateway automatically intercepts the interaction and escalates required assurances from *Claim/Brief* to *Proof + Stake + Hard Constraint*, treating Reputation strictly as a routing and prioritization overlay rather than a single security gate.

VI. Attacks, Vulnerabilities, and Defense Mechanisms in Inter-Agent Trust

Because trust management systems directly govern partner selection, resource access, and message acceptance, the trust plane itself is a primary target for adversarial manipulation [18], [33], [83], [94]. Threats to inter-agent trust fall into two overarching categories: (1) *classical reputation and identity attacks* that manipulate rating aggregation or exploit cheap pseudonyms, and (2) *agentic and LLM-MAS interaction attacks* that weaponize natural-language persuasion, communication interception, and contagious prompt propagation. Table VII synthesizes the complete attack-defense matrix.

A. Classical Reputation, Identity, and Strategic Timing Attacks

In decentralized P2P, IoT, vehicular, and marketplace MAS, malicious agents attempt to subvert trust computation through five canonical attack vectors [14], [15], [18], [33], [38]:

Sybil Attack: A single adversary fabricates a massive number of pseudonymous identities $\mathcal{S}_{\text{sybil}}$ to dominate voting quorums, outvote honest witnesses, or partition DHT overlays [15], [27].

Whitewashing and Cheap Pseudonyms: After accumulating a high disbelief score $d_j^i \to 1$ through defection, a malicious agent discards its tarnished identity and re-enters the system under a fresh identity at the neutral cold-start prior T_0 [14].

Bad-Mouthing (Slandering) and Ballot-Stuffing: In bad-mouthing, colluding advisors submit false negative ratings to depress the reputation of honest competitors; in ballot-stuffing,

Table VI
COMPARISON OF INTER-AGENT COMMUNICATION AND TRUST PROTOCOLS ACROSS THE SIX TRUST MODELS.

Protocol / Standard	Primary Scope	Claim	Brief	Proof	Stake	Rep.	Constr.	Key Architectural Mechanism	Refs.
Anthropic MCP	Agent-to-Tool / Context	●	◐	○	○	○	●	Standardizes JSON-RPC tool/context access; relies on host sandboxing	[84]
Google A2A	Enterprise Inter-Agent	●	◐	○	○	○	●	AgentCard capability discovery + HTTPS/OAuth/mTLS secure messaging	[86], [94]
ANP & BlockA2A	Cross-Domain DID-A2A	●	●	◐	○	◐	◐	Binds A2A AgentCards to W3C DIDs (did:wba / DLT) with Ed25519 signatures	[84], [92]
Google/PayPal AP2	Agentic Commerce	◐	●	●	◐	◐	●	Cryptographic signed user Mandates + payment tokenization & role separation	[87], [94]
A2A + HTTP x402	DeFi / Micropayments	●	●	●	◐	◐	◐	Smart-contract AgentCard registry + HTTP 402 atomic per-call settlement	[91]
Ethereum ERC-8004	Trustless Web3 Agents	●	◐	●	●	●	◐	Composable on-chain Identity (NFT), Reputation, and Validation (TEE/ZKP/Stake) registries	[88], [94]
NANDA Index	Global Agent Discovery	●	●	◐	○	◐	◐	Decentralized index of Verified AgentFacts & third-party safety attestations	[89]
Eclipse LMOS	Internet of Agents (IoA)	●	●	◐	○	◐	●	3-layer architecture (JSON-LD Application, Transport, W3C DID/OAuth2 Security)	[84]

● Native protocol primitive. ◐ Supported via optional extensions or off-path hooks. ○ Not specified by protocol.

colluders execute wash transactions to artificially inflate the reputation of a malicious accomplice [25], [28].

Strategic On-Off (Oscillating) and Reputation Milking Attacks: A rational adversary behaves cooperatively across numerous low-value transactions to build high trust ($T_{i,j} \to 1$), then strategically defects on a high-value transaction or alternates between cooperative and defective phases just above the eviction threshold [32], [94].

Conflicting Behavior (Discrimination) Attack: A malicious trustee provides exemplary service to a majority of peers (or pre-trusted power nodes) while selectively defecting against a targeted minority victim, causing global spectral algorithms to dismiss the victim's complaints as outlier noise [33], [38].

B. Agentic and LLM-MAS Interaction Attacks

In foundation-model multi-agent systems, attackers exploit both the semantic openness of natural language and the graph topology of inter-agent workflows [83], [94], [102]:

Cooperative Persuasion and Debate Manipulation Attacks: Multiple compromised or adversarial agents coordinate within a multi-agent debate to overwhelm honest agents with fabricated evidence, exploiting LLM sycophancy and nudge-susceptibility to flip consensus toward counterfactual or unsafe conclusions [58]–[60].

Agent-in-the-Middle (AiTM) Communication Interception: Analogous to network man-in-the-middle attacks, an intermediary routing or orchestrator agent intercepts messages between planner and executor agents, subtly rewriting task constraints or injecting malicious tool parameters while preserving linguistic fluency [80].

Infectious Prompt Contagion (Prompt Infection, CORBA, and Agent Smith): Unlike single-agent prompt injection, *Prompt Infection* [56] crafts self-replicating adversarial payloads that instruct a compromised LLM agent to append the same payload to all outbound messages sent to neighboring peers. *CORBA (Contagious Recursive Blocking Attack)* [81] introduces topology-independent recursive loops that propagate across MAS networks to exhaust token budgets and block task completion, while *Agent Smith* [57] extends infectious contagion to multimodal image-text inputs.

Shared Memory Poisoning and Confused-Deputy Tool Escalation: A low-privilege agent writes poisoned entries into a shared vector RAG database or manipulates an A2A/MCP request so that a high-privilege peer agent executes an unauthorized action using its own elevated credentials [83], [94].

C. Defense-in-Depth Countermeasures and Evaluation Benchmarks

Mitigating the full spectrum of classical and agentic attacks requires a layered defense-in-depth architecture combining five countermeasure families: (1) **Cryptographic Identity and Economic Entry Barriers** (binding agents to W3C DIDs/VCs, requiring non-refundable entry fees or ERC-8004 collateral stakes, and signing every message via BlockA2A/AP2 Mandates to eliminate Sybil, whitewashing, and AiTM spoofing) [88], [92], [93]; (2) **Robust Statistical Filtering and Asymmetric Decay** (applying TRAVOS beta-deviation filtering, PeerTrust rater credibility weighting, and fast-decay penalty multipliers against bad-mouthing, ballot-stuffing, and on-off milking) [25], [28], [32]; (3) **Internal Neural Screening (A-Trust TMS)** (inspecting multi-head attention signatures across the six Gricean dimensions to catch subtle misinformation and prompt injections before context assimilation) [102]; (4) **Topological & Consensus Defenses** (deploying G-Safeguard GNN discourse monitors [82] and BlockAgents Proof-of-Thought debate consensus [62] to quarantine infectious CORBA/Prompt Infection nodes); and (5) **Formal Runtime Constraints** (enforcing container sandboxing and Linear Temporal Logic (LTL) action pruning such as SELP [66], [67]). Standardized evaluation benchmarks—including *SafeAgentBench* [69], *Agent-SafetyBench* [70], *Agent Security Bench (ASB)* [71], *R-Judge* [72], and the *Trust Viola-*

Table VII
COMPREHENSIVE ATTACK-DEFENSE MATRIX FOR INTER-AGENT TRUST ACROSS CLASSICAL AND FOUNDATION-MODEL MULTI-AGENT SYSTEMS.

Attack Vector	Target Layer	Exploited Vulnerability	Primary Defense Mechanisms	Representative Schemes	Refs.
Sybil & Whitewashing	Identity & Registry Layer	Cheap pseudonyms; resetting negative reputation to neutral prior T_0	W3C DIDs + third-party VCs; ERC-8004 staking bonds; pre-trusted EigenTrust anchors	ERC-8004, BlockA2A, EigenTrust	[14], [15], [27], [88]
Bad-Mouthing & Ballot-Stuffing	Witness Aggregation Layer	Collusive false ratings inflating accomplices or slandering honest peers	TRAVOS accuracy binning; PeerTrust PSM similarity credibility; ATTR weighting	TRAVOS, PeerTrust, BRS	[25], [28], [43]
On-Off & Reputation Milking	Temporal Trust Dynamics	Accumulating high reputation on small tasks to defect on high-stakes tasks	Asymmetric fast-destruction decay; SecuredTrust load/deviation tracking; Tier T2/T3 Proof+Stake gating	SecuredTrust, Tiered T0--T3	[32], [94]
Cooperative Persuasion & Sycophancy	LLM-MAS Debate Layer	Adversarial peer nudges exploiting RLHF sycophancy to flip consensus	Multi-round Proof-of-Thought (PoT) voting; Audit-LLM atomic claim verification; A-Trust bias/logic heads	BlockAgents, Audit-LLM, A-Trust	[59], [62], [76], [102]
Agent-in-the-Middle (AiTM)	Inter-Agent Transport	Intermediary agent intercepting and mutating routed prompts/commands	End-to-end DID message signing (SG-A2A/VG-A2A); co-signed AP2 user intent Mandates	BlockA2A, AP2, ANP	[80], [87], [92]
Infectious Contagion (Prompt Infection / CORBA)	MAS Discourse Graph	Self-replicating prompt/multimodal payloads spreading across agent topologies	G-Safeguard GNN anomaly pruning; GPTSwarm edge optimization; A-Trust Quarantine policy; circuit breakers	G-Safeguard, GPTSwarm, A-Trust	[56], [57], [81], [82]
Confused-Deputy & Tool/Memory Poisoning	Execution & RAG Layer	Subverting a privileged peer into executing unauthorized API/physical actions	Scoped ephemeral tokens; container sandboxes; LTL formal safety pruning (SELP); TEE/zkML attestation	SELP, TrustAgent, TRiSM	[67], [83], [85]

tion Dataset [102]—provide reproducible testbeds for stress-testing these defenses.

VII. APPLICATIONS AND USE CASES

Inter-agent trust mechanisms underpin a broad spectrum of cyber-physical, wireless networking, and digital ecosystems. Following the application-centric methodology of IEEE COMST surveys, we examine eight representative deployment scenarios where inter-agent trust directly governs system safety, spectral/computational efficiency, and economic viability.

1) Internet of Things (IoT) and Social IoT (SIoT) Service Management: In large-scale IoT and Social IoT (SIoT) deployments, billions of heterogeneous, resource-constrained devices autonomously discover and compose sensing and actuation services [38], [39]. Because centralized cloud trust registries suffer from prohibitive signaling latency and single-point failures, hierarchical fog/cloudlet architectures (such as BC-Trust and mobile cloud IoT trust protocols [39]) combine direct packet-delivery and data-consistency observations with social similarity metrics (co-location, co-work, and parental object relationships) to isolate malfunctioning or compromised sensors while minimizing energy overhead.

2) Internet of Vehicles (IoV), V2X, and Connected Autonomous Vehicles (CAVs): In V2X and CAV networks, vehicles exchange safety-critical Basic Safety Messages (BSMs), cooperative perception maps, and platooning maneuvers at millisecond timescales under high mobility [68], [98]. While vehicular PKI certificates authenticate external identities, internal compromised vehicles can broadcast fabricated hazard warnings. Modern IoV trust frameworks therefore deploy *hybrid node-centric and data-centric trust models* [98]: data-centric modules evaluate the physical plausibility and spatial-temporal consistency of reported traffic events (e.g., via SSTM neighbor voting [43] or causal perception reasoning in Hudson [83]), while node-centric reputation and roadside-unit (RSU) consortium blockchains track long-term vehicle reliability.

3) Multi-UAV Swarms and Space-Air-Ground Integrated Networks (SAGIN): Autonomous multi-UAV swarms and SAGIN architectures coordinate aerial base stations, low-earth-orbit (LEO) satellites, and terrestrial gateways for disaster recovery, cooperative target tracking, and 6G coverage extension [48], [73]. Operating in contested, GPS-denied, or intermittent-connectivity environments exposes UAV swarms to Byzantine relay dropping, false spectrum sensing reports, and trajectory spoofing. Trust-aware MARL and subjective-logic path discounting enable UAVs to dynamically select trustworthy relay hops and form resilient task coalitions without relying on a ground control station [36], [73].

4) 6G Wireless Resource Management and Federated Multi-Agent Learning: In next-generation 6G networks, autonomous network slice brokers, Radio Access Network (RAN) intelligent controllers, and edge nodes dynamically share spectrum, infrastructure, and compute via Blockchain Radio Access Networks (B-RAN) and decentralized auctions [42], [74]. When multiple edge agents collaboratively train foundation or control models via Federated Learning (FL) and Decentralized MARL, malicious participants can launch model-poisoning or free-riding attacks. Integrating reputation registries with zero-knowledge gradient proofs and stake slashing ensures that only verified, high-trust edge agents contribute to global policy aggregation.

5) Autonomous Agentic Commerce, DeFi, and Supply Chain Negotiation: In the emerging Agentic Web, personal shopping agents, enterprise procurement agents, and decentralized finance (DeFi) arbitrage agents autonomously discover counterparties via A2A/NANDA, negotiate multi-issue supply contracts, and settle transactions via AP2 Mandates or HTTP x402 micropayments [87]–[89], [91]. Here, Tier T1 and T2 hybrid trust architectures—combining W3C DIDs/VCs for corporate authority verification, ERC-8004 Reputation

Registries for partner selection, and cryptographically signed spending Mandates—prevent unauthorized budget depletion, price-gouging manipulation, and counterparty default.

6) Collaborative Software Engineering and Automated Scientific Discovery: Multi-agent coding and research frameworks—such as MetaGPT [50] and AutoGen [49]—orchestrate specialized product manager, architect, coder, and QA tester agents to synthesize complex software repositories and scientific simulations. Without inter-agent trust verification, a hallucinated API specification or an indirect prompt injection hidden in an external documentation page can propagate through the coder-reviewer loop. Deploying A-Trust attention screening [102] alongside containerized test-execution Proofs and G-Safeguard discourse monitoring [82] substantially improves compilation accuracy and blocks vulnerable code injection.

7) Clinical Healthcare Decision Support and Privacy-Critical Ecosystems: In multi-agent clinical workflows (such as Polaris and knowledge-driven medical MAS [83], [85]), specialist diagnostic, pharmacology, and triage agents collaborate over sensitive electronic health records (EHRs). Under the Tier T3 trust regime, inter-agent interactions require strict regulatory Briefs (clinical software certifications), context desensitization and zero-knowledge privacy proofs to prevent HIPAA/GDPR data leakage across agents, and mandatory human-in-the-loop physician sign-off before treatment actuation.

8) Embodied Robotics and Industrial Cyber-Physical Control: In smart manufacturing, programmable logic controller (PLC) automation (e.g., Agents4PLC [83]), and multi-robot warehouse teams, LLM-based planner agents issue task commands to physical robot actuators. To prevent catastrophic physical harm from hallucinated or compromised inter-agent plans, frameworks such as SELP [67] and LTL constraint pruning [66] combine equivalence voting across multiple planner agents with formal Linear Temporal Logic model checking, guaranteeing that only provably safe action trajectories reach the hardware controller.

VIII. Lessons Learned, Open Challenges, and Future Research Directions

Our comprehensive synthesis of theoretical definitions, mathematical models, agentic protocols, and domain applications reveals that while inter-agent trust has matured significantly over three decades, the convergence of foundation models, decentralized ledgers, and real-time cyber-physical networks exposes six fundamental research challenges, summarized in Fig. 5.

A. The Verification-Latency Paradox in Real-Time Inter-Agent Coordination

A primary lesson from comparing mathematical models (Section III) and verifiable protocols (Section V) is the acute trade-off between verification assurance and interaction latency. Cryptographic proofs (such as zkML circuit proving and on-chain ERC-8004 validation) and multi-round LLM debate consensus provide strong security against deception, yet incur seconds to minutes of computational overhead and financial gas costs. Conversely, ultra-low-latency applications—such as 6G dynamic spectrum access, CAV platooning, and high-frequency agentic microservice calls—require sub-10-millisecond trust decisions. Future research must pioneer *speculative and optimistic trust architectures*: executing low- and moderate-risk interactions immediately via zero-overhead internal neural checks (such as A-Trust multi-head attention scoring [102] and cached DID session keys), while asynchronously verifying randomized batches via zero-knowledge proofs and slashing bonded stake ex-post if discrepancies are uncovered.

B. Decoupling Historical Reputation from Runtime Cognitive Hijacking

Classical trust literature implicitly assumes that an agent's past honesty strongly predicts its immediate future behavior. In foundation-model MAS, however, an agent with a flawless historical reputation can be hijacked mid-dialogue via indirect prompt infection [56] or nudged into sycophantic agreement by an adversarial peer [76], [94]. Treating reputation as a static passkey is therefore a critical design anti-pattern. Future trust frameworks must tightly integrate *longitudinal entity reputation* with *in-situ message-level and activation-level anomaly detection*, automatically resetting an agent's effective trust tier to T0 whenever its internal attention entropy, discourse graph trajectory, or model weights undergo a distribution shift.

C. Cross-Domain Trust Transferability and Semantic Protocol Interoperability

As the ecosystem fragments across competing standards—Google A2A, Anthropic MCP, ANP, NEAR AITP, Cisco AConP, Eclipse LMOS, and Ethereum ERC-8004 [84], [94]—an agent operating in an enterprise A2A environment cannot seamlessly interpret the trust standing of a Web3 ERC-8004 agent or a 6G SIoT device using Subjective Logic opinions. Developing formal *trust homomorphisms and semantic translation bridges*—grounded in the bijective mappings between Dirichlet/Beta evidence spaces and Subjective Logic opinions [19], [43], and standardized through W3C Verifiable Credential schemas—is an urgent priority for cross-ecosystem interoperability.

D. Mitigating Emergent Steganographic Collusion in Foundation-Model Societies

Traditional anti-collusion mechanisms (such as EigenTrust pre-trusted peers and PeerTrust similarity weighting [27], [28]) detect explicit rating cliques. However, generative LLM agents capable of multi-turn natural language communication can spontaneously learn or be prompted to use *linguistic steganography*—encoding hidden coordination signals within benign-looking prose or code comments—to coordinate market manipulation, price fixing, or synchronized ballot-stuffing without triggering surface-level content filters. Countering steganographic inter-agent collusion requires combining information-theoretic communication channel sanitization (e.g., deterministic semantic canonicalization or paraphrasing gateways between untrusted agents) with game-theoretic peer-prediction mechanisms where truth-telling is a strict dominant strategy.

Core Engineering Tension / Pitfall	Root Technical Cause	Future Research Direction
1. Verification-Latency Paradox	High-assurance Proofs (zkML, TEE attestation, on-chain consensus) require hundreds of ms to seconds, clashing with sub-10ms V2X/6G/agentic loops.	Speculative optimistic execution with asynchronous zk-audits, lightweight internal A-Trust screening, and hardware-accelerated zk-ASICs.
2. Runtime Cognitive Nudge Fragility	Classical reputation assumes stationary competence, whereas RLHF LLMs can be socially engineered mid-conversation via sycophancy or prompt infection.	Coupling static reputation with continuous in-situ attention/activation monitoring and non-accumulating zero-trust action gates.
3. Cross-Protocol Semantic Fragmentation	Heterogeneous trust representations across A2A, MCP, ANP, ERC-8004, Subjective Logic, and BDTM hinder cross-domain trust transfer.	Standardized W3C/IETF/IEEE ontology translators and Generic Bayesian / Subjective Logic homomorphic trust bridges.
4. Steganographic Multi-Agent Collusion	Foundation-model agents can encode covert coordination signals in natural language syntax, evading statistical outlier and GNN detectors.	Information-theoretic channel capacity bounding, randomized semantic paraphrasing proxies, and mechanism-design collusion-proof scoring rules.
5. Privacy vs. Auditability Dilemma	Global reputation ledgers and signed action receipts expose proprietary workflows and user behavior to inference attacks.	Zero-knowledge reputation proofs (proving $T_{i,j} \geq \tau$ without revealing history) and differentially private federated trust aggregation.
6. Lack of Longitudinal Dynamic Benchmarks	Existing benchmarks test static single-turn jailbreaks rather than long-horizon strategic sleeper agents and multi-hop economic collusion.	Open, standardized multi-agent economic and cyber-physical arena benchmarks with adaptive red-team sleeper populations.

Figure 5. Summary of key lessons learned, root technical bottlenecks, and future research directions in inter-agent trust.

E. Reconciling Privacy-Preserving Trust with Regulatory Auditability

Publicly readable reputation registries and immutable audit logs facilitate transparency and accountability, yet they risk leaking sensitive commercial relationships, proprietary tool-invocation patterns, and personal user intents [85], [94]. Future research should advance *Zero-Knowledge Reputation and Selective-Disclosure Credentials* (e.g., BBS+ signatures and zk-SNARKs over verifiable credentials), enabling an autonomous agent to prove that its reputation score exceeds a threshold ($T_{i,j} \geq \tau$), that its stake bond is funded ($S_j \geq S_{\min}$), and that it holds a valid regulatory license without revealing its underlying transaction history or principal identity except under authorized regulatory escrow.

F. Standardized Longitudinal Benchmarks for Dynamic Trust Calibration

While benchmarks such as SafeAgentBench [69], AgentSafetyBench [70], ASB [71], and the Trust Violation Dataset [102] represent important milestones, most current evaluations measure static, few-turn attack success rates. There remains a pressing need for open, longitudinal multi-agent simulation arenas that evaluate *dynamic trust calibration accuracy*—quantifying both over-trust errors and under-trust opportunity costs—across thousands of episodes involving strategic sleeper agents, evolving task distributions, network churn, and hybrid protocol costs.

IX. Conclusion

In this paper, we presented a comprehensive survey of trust in inter-agent interactions, unifying three decades of foundational multi-agent systems research with contemporary advances in 6G/IoT network engineering, decentralized cryptography, and foundation-model Agentic AI. We began by tracing the conceptual evolution of inter-agent trust across sociological, socio-cognitive, computational, probabilistic, and linguistic-pragmatic dimensions, establishing six formal mathematical axioms and a six-dimensional taxonomy that clearly distinguishes trust from reputation, trustworthiness, reliability, and hard security. We then formulated and comparatively analyzed six core families of mathematical trust models—spanning Generic Bayesian sequential inference (BDTM, DDTM, TRAVOS, SSTM), Subjective Logic and fuzzy/possibilistic reasoning, socio-cognitive composite architectures (ReGreT, FIRE), spectral graph algorithms (EigenTrust, PeerTrust, PowerTrust), game-theoretic and crypto-economic stake-slashing mechanisms, and machine learning approaches including MARL and internal multi-head attention trust scoring (A-Trust).

Furthermore, we examined the profound structural shift introduced by LLM-based multi-agent systems, demonstrating why cognitive fragilities such as prompt infection, sycophancy, hallucination cascades, sleeper deception, and insider misalignment render standalone claim or reputation schemes insufficient. Through an in-depth evaluation of emerging inter-agent protocols—MCP, Google A2A, ANP, BlockA2A, AP2, HTTP x402, Ethereum ERC-8004, W3C DIDs/VCs, NANDA, zkML, and TEE remote attestations—we synthesized a risk-calibrated Tiered Hybrid Trust Blueprint (T0–T3) that pairs low-friction discovery for routine tasks with defense-in-depth cryptographic verification, economic collateral, and sandboxed constraints for high-stakes actions. Finally, by cataloging the complete attack-defense matrix, reviewing eight cyber-physical and digital application domains, and distilling six open research challenges, this survey provides researchers and practitioners with a rigorous roadmap for engineering resilient, accountable, and trustworthy multi-agent societies.

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